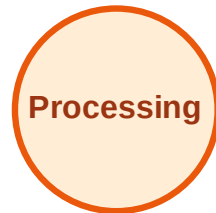


BFS Traversal

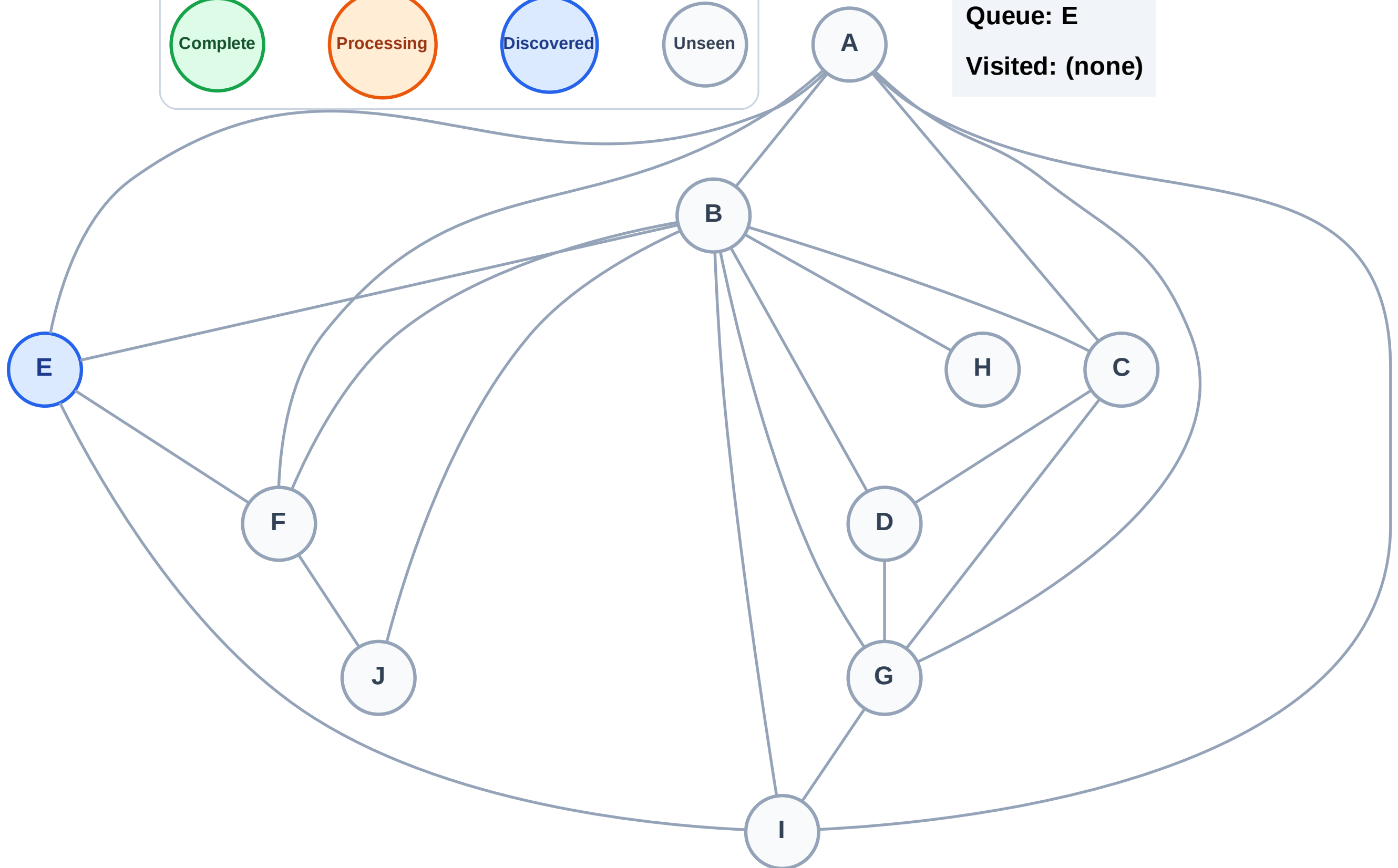
Step 1: Start at E

Legend



Queue: E

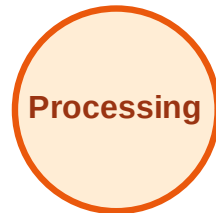
Visited: (none)



BFS Traversal

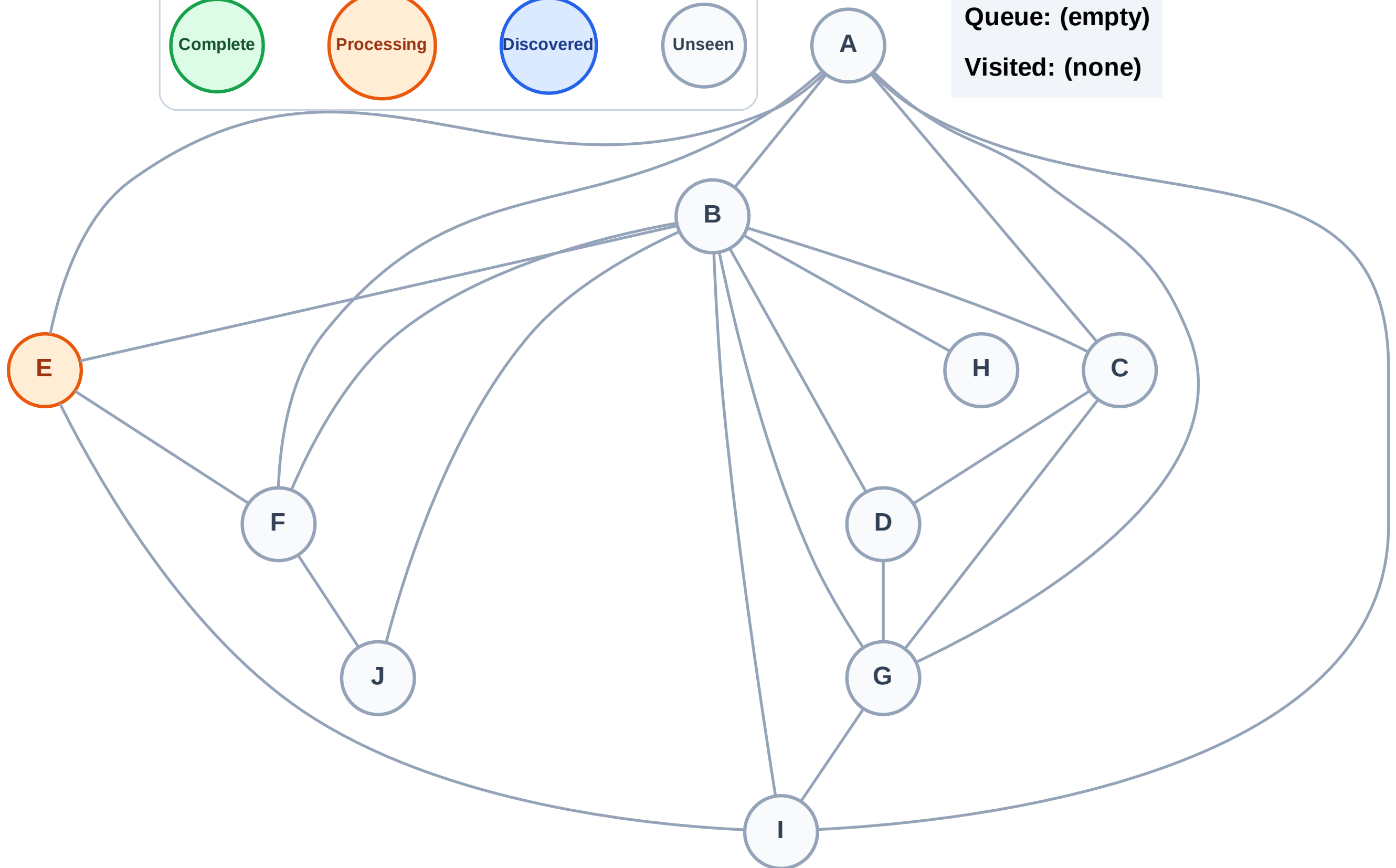
Step 2: Process node E

Legend



Queue: (empty)

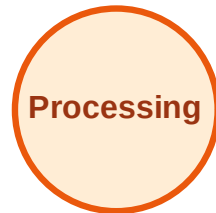
Visited: (none)



BFS Traversal

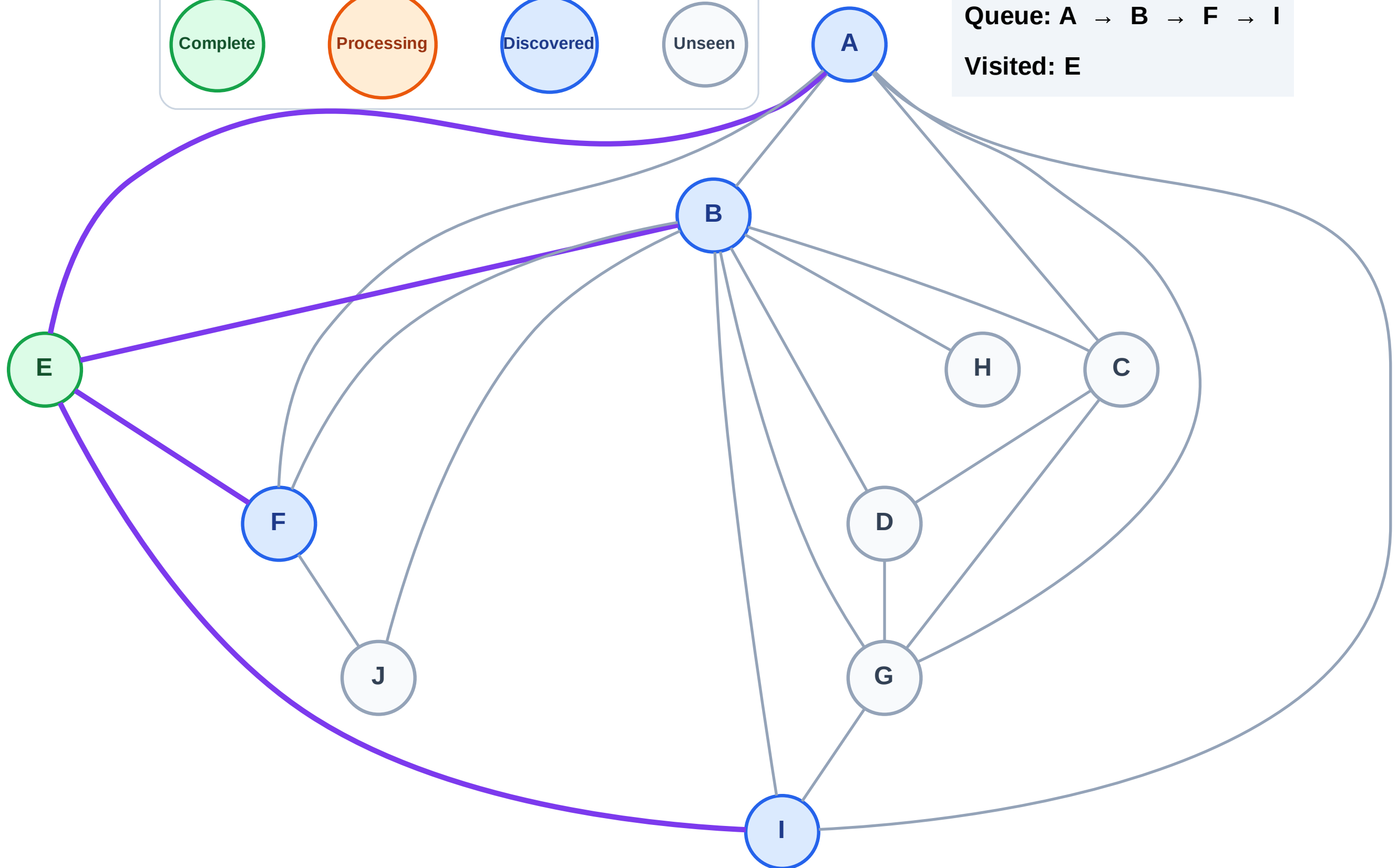
Step 3: Discover A, B, F, I from E

Legend



Queue: A → B → F → I

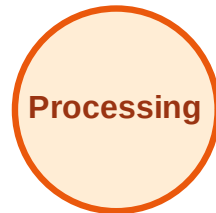
Visited: E



BFS Traversal

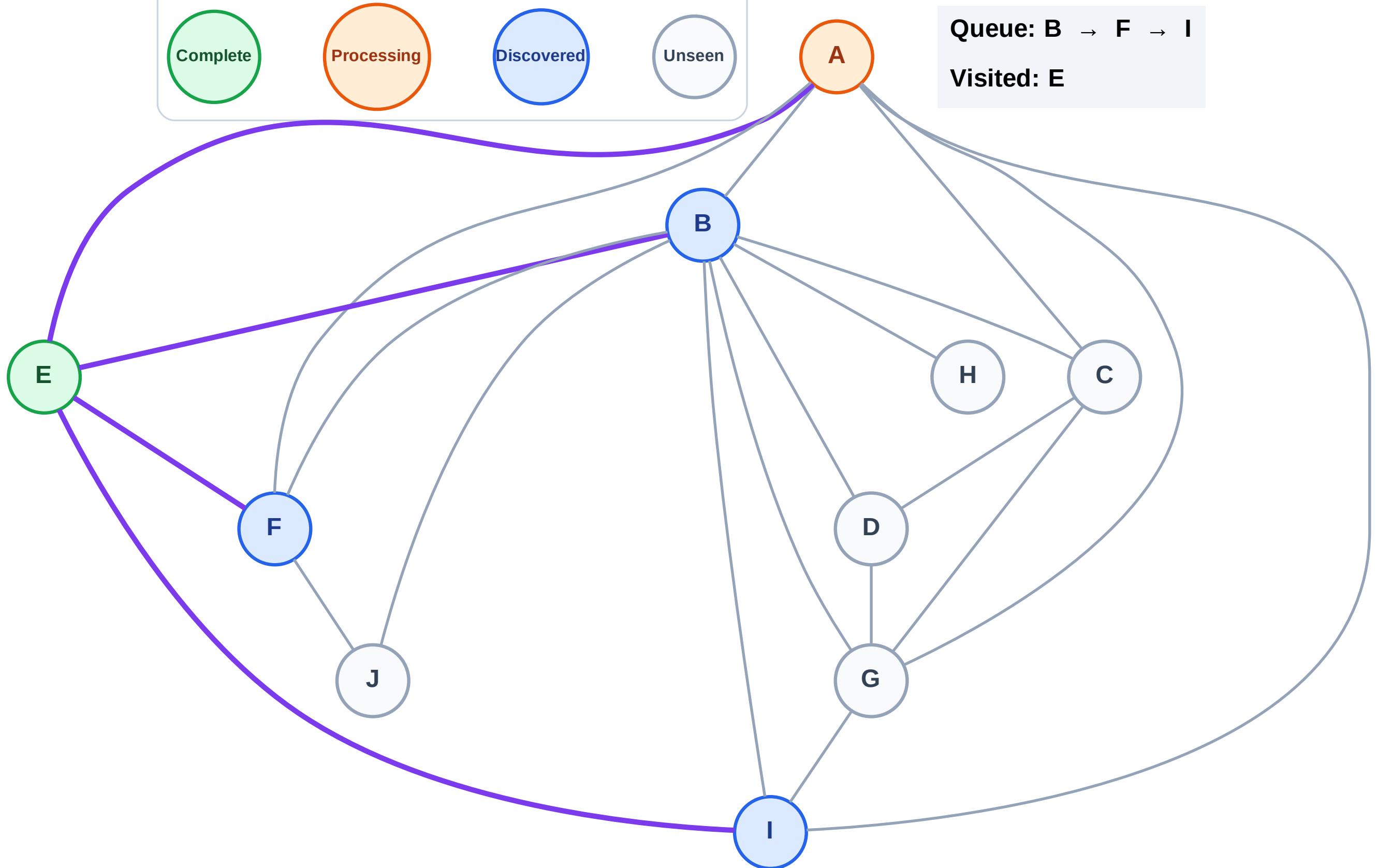
Step 4: Process node A

Legend



Queue: B → F → I

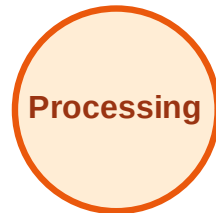
Visited: E



BFS Traversal

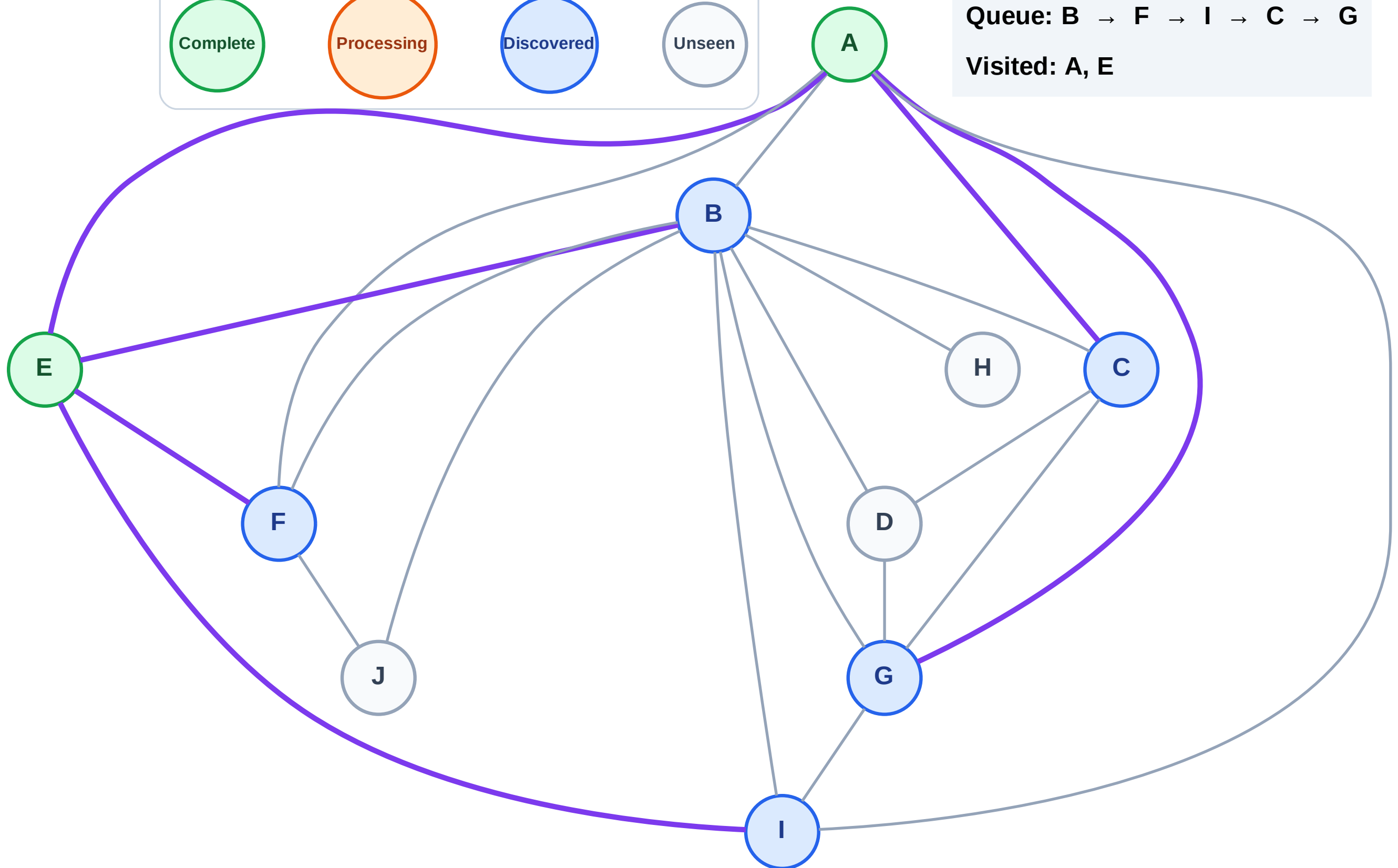
Step 5: Discover C, G from A

Legend



Queue: B → F → I → C → G

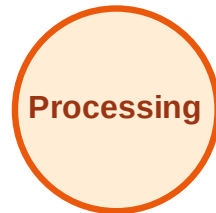
Visited: A, E



BFS Traversal

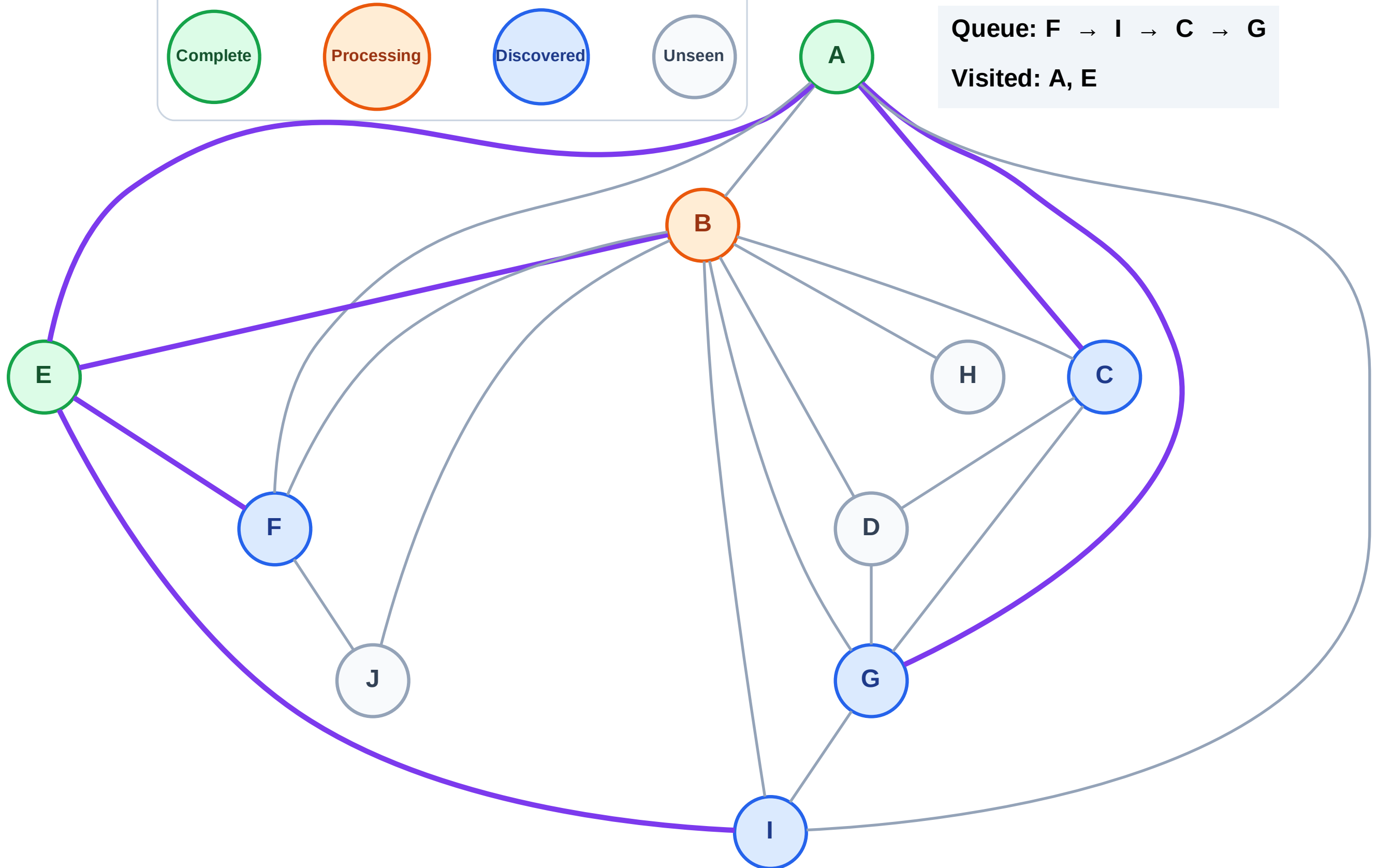
Step 6: Process node B

Legend



Queue: F → I → C → G

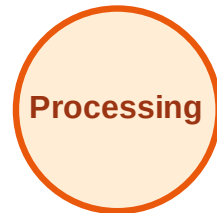
Visited: A, E



BFS Traversal

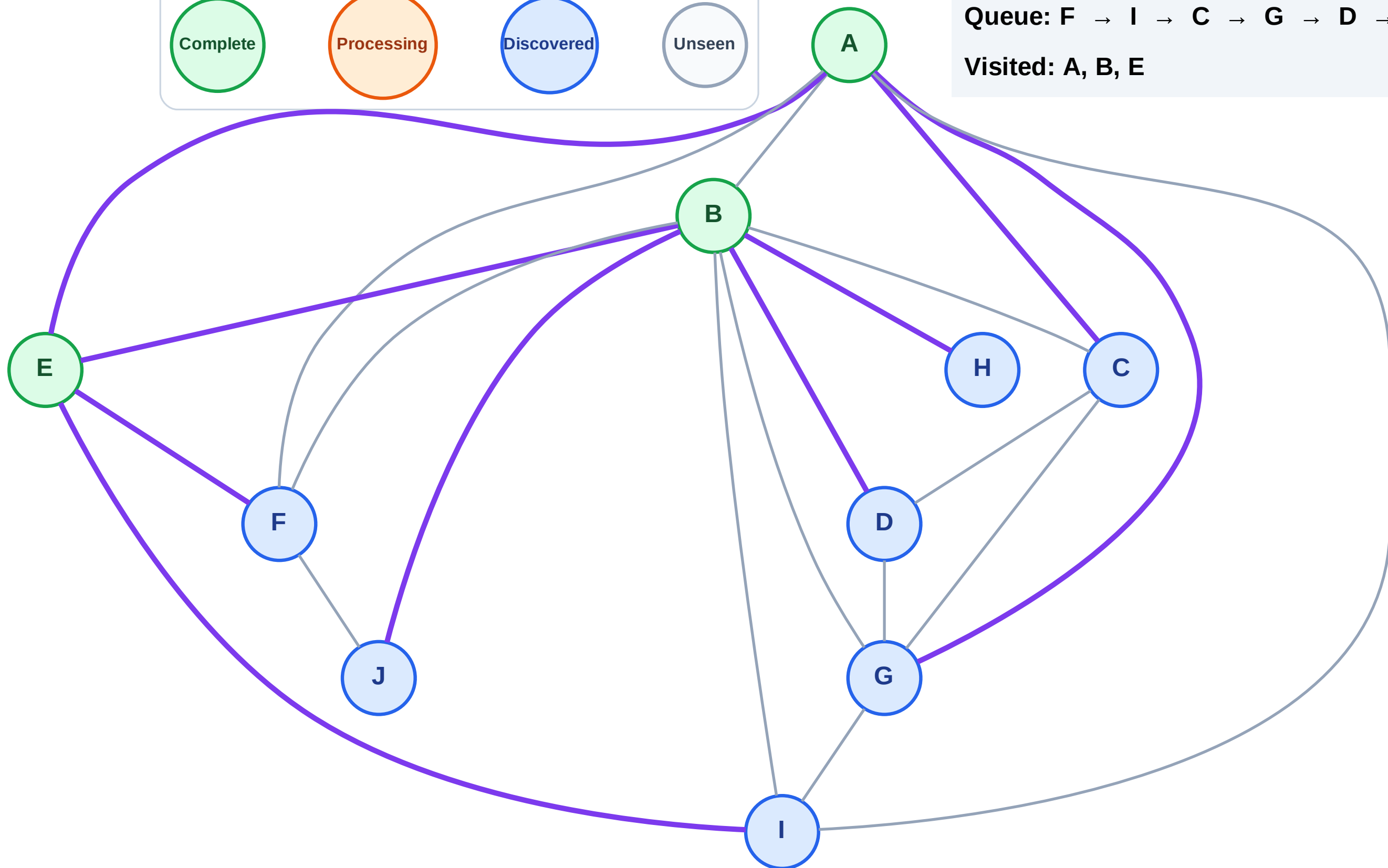
Step 7: Discover D, H, J from B

Legend



Queue: F → I → C → G → D → H → J

Visited: A, B, E



BFS Traversal

Step 8: Process node F

Legend

Complete

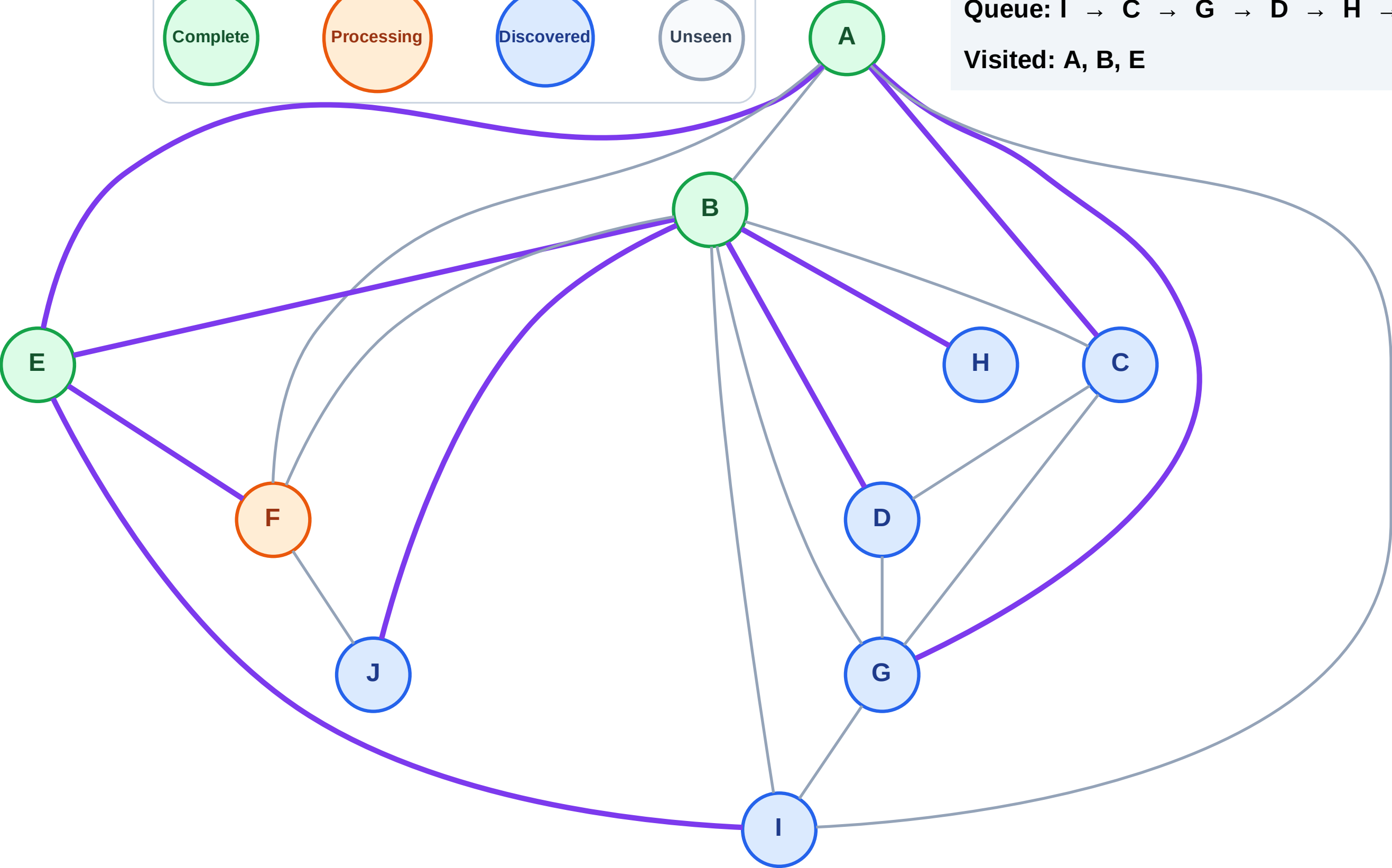
Processing

Discovered

Unseen

Queue: I → C → G → D → H → J

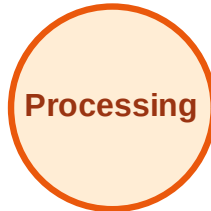
Visited: A, B, E



BFS Traversal

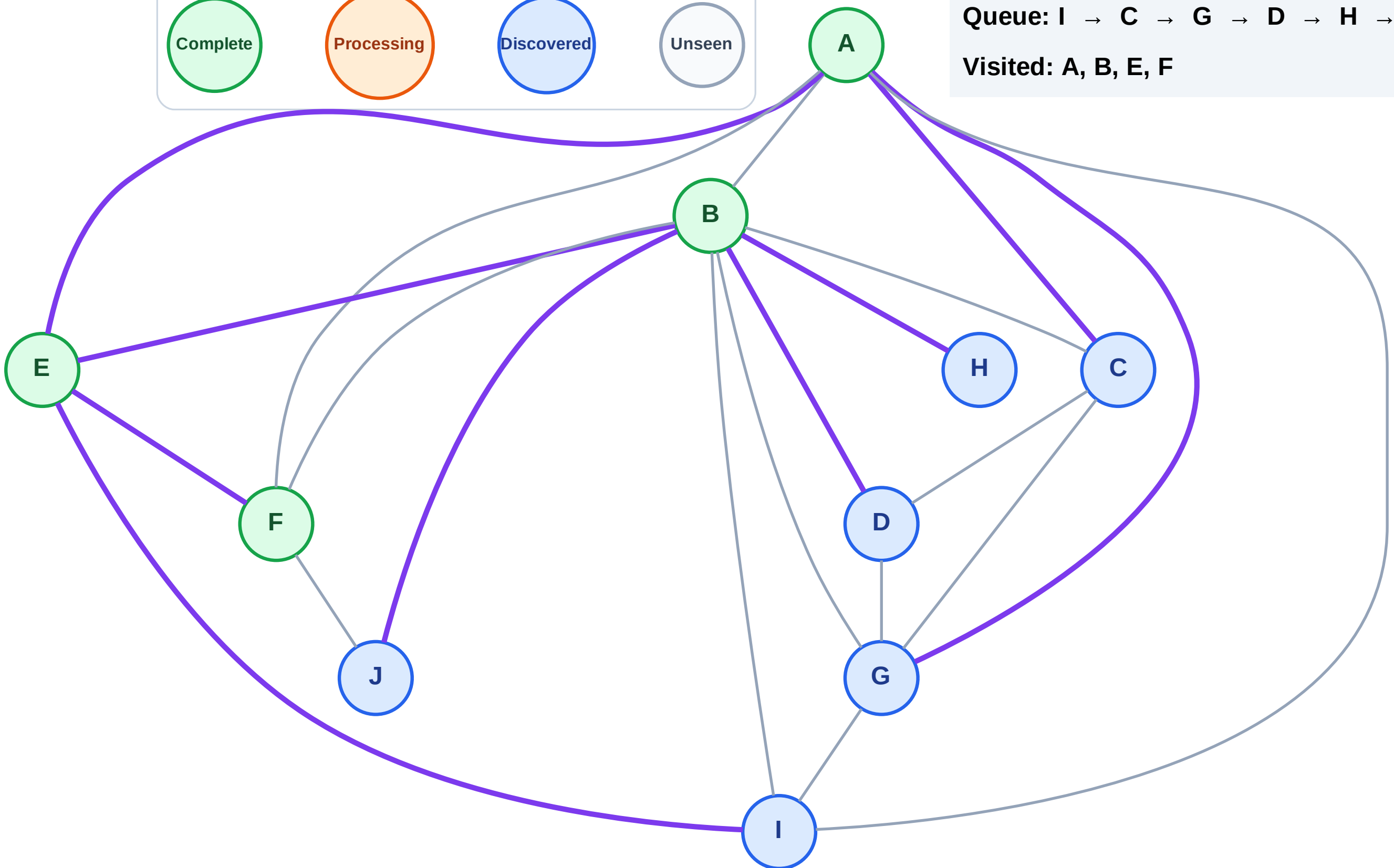
Step 9: No new neighbors from F

Legend



Queue: I → C → G → D → H → J

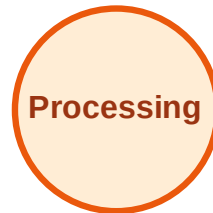
Visited: A, B, E, F



BFS Traversal

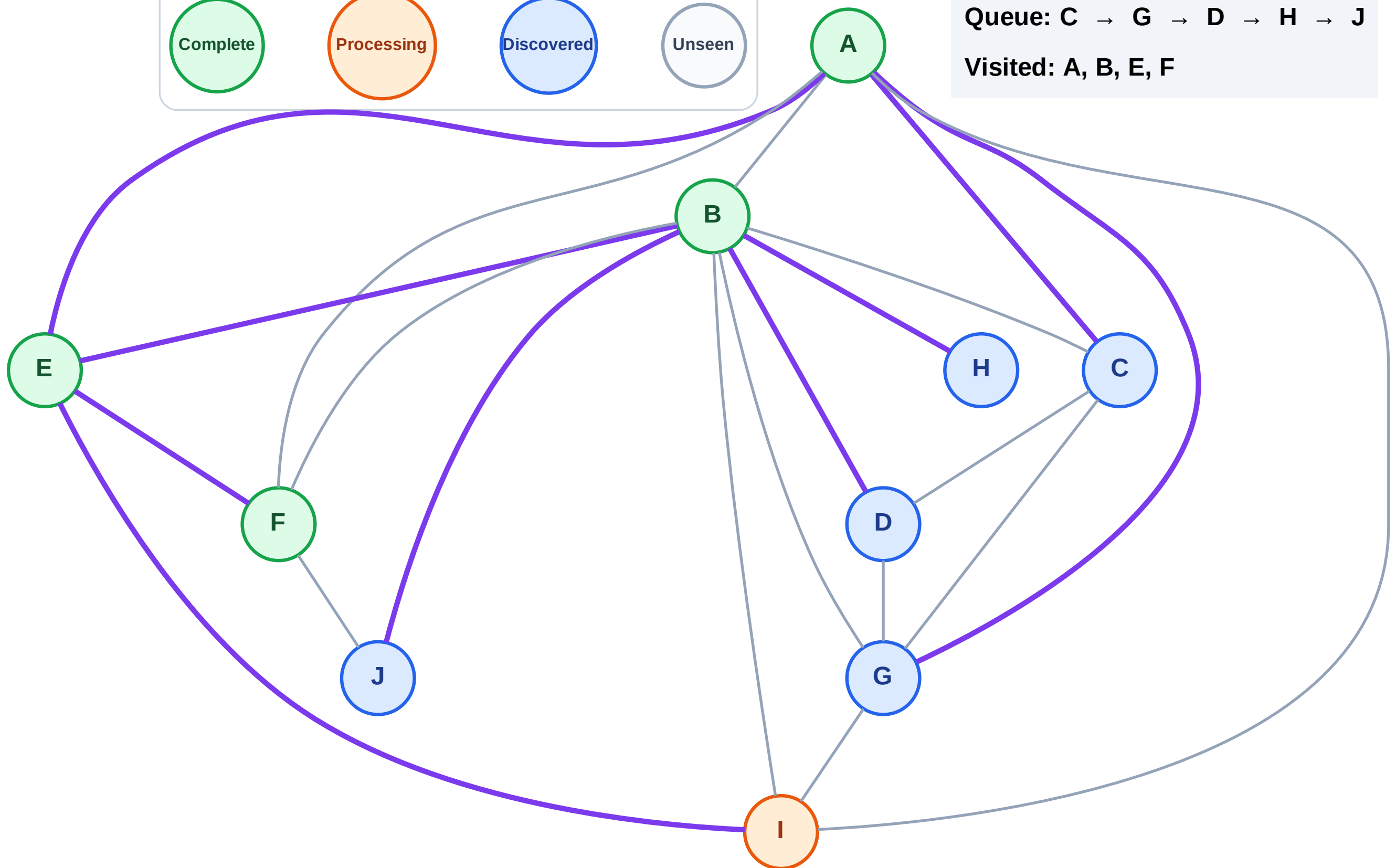
Step 10: Process node I

Legend



Queue: C → G → D → H → J

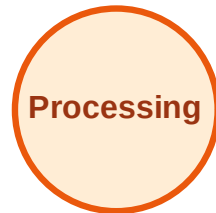
Visited: A, B, E, F



BFS Traversal

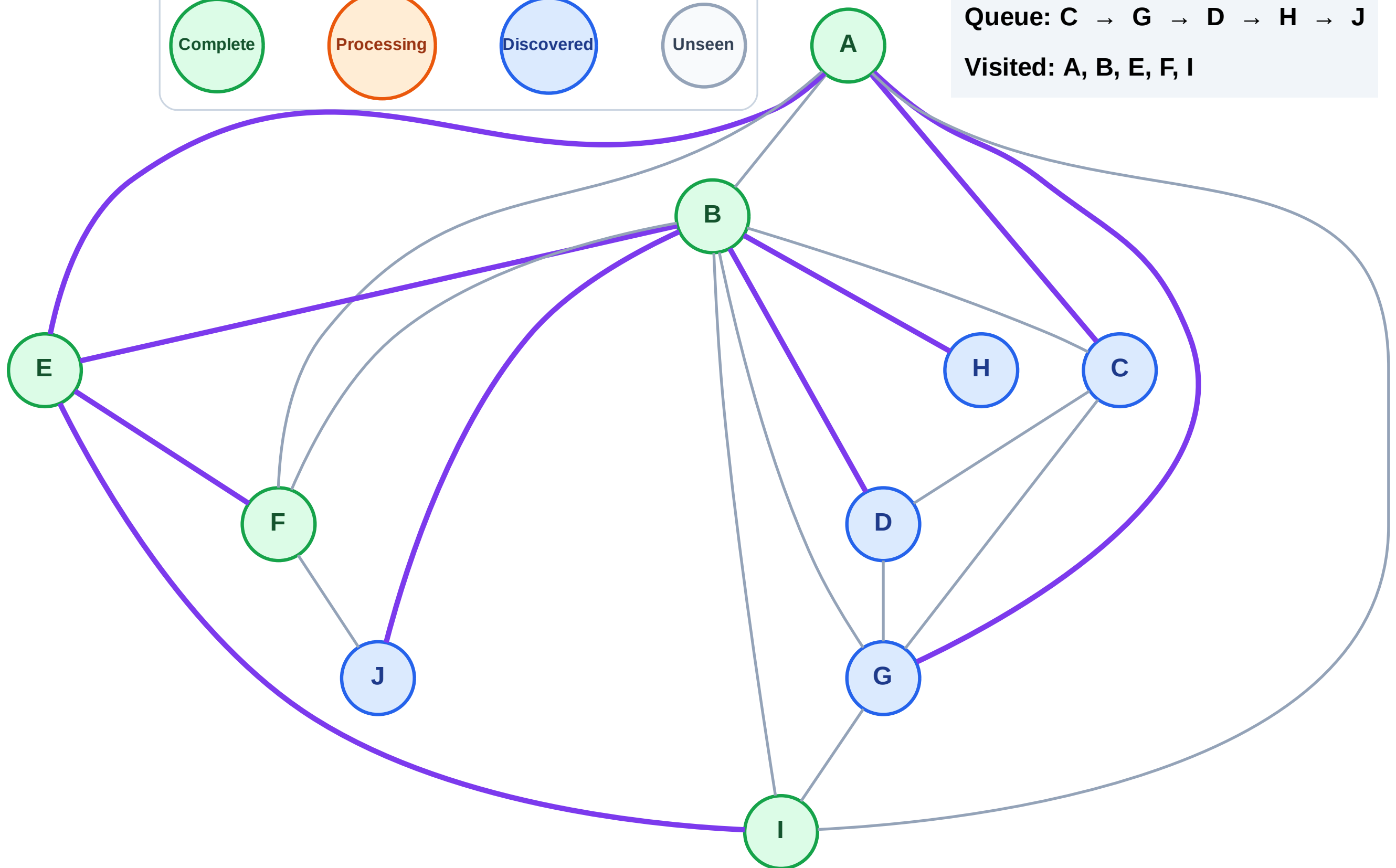
Step 11: No new neighbors from I

Legend



Queue: C → G → D → H → J

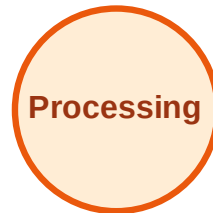
Visited: A, B, E, F, I



BFS Traversal

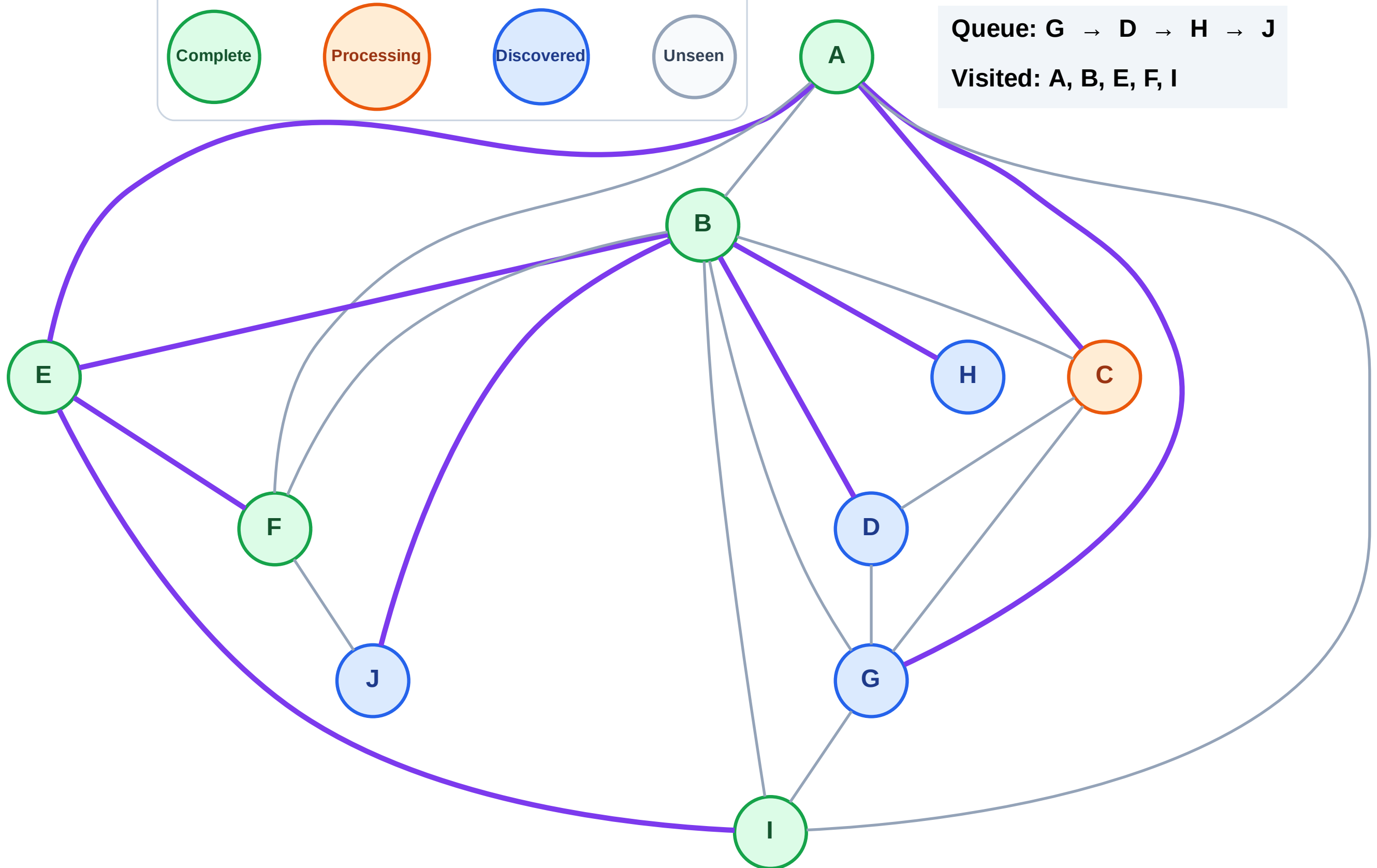
Step 12: Process node C

Legend



Queue: G → D → H → J

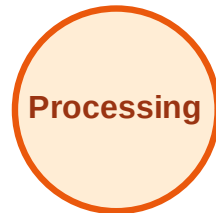
Visited: A, B, E, F, I



BFS Traversal

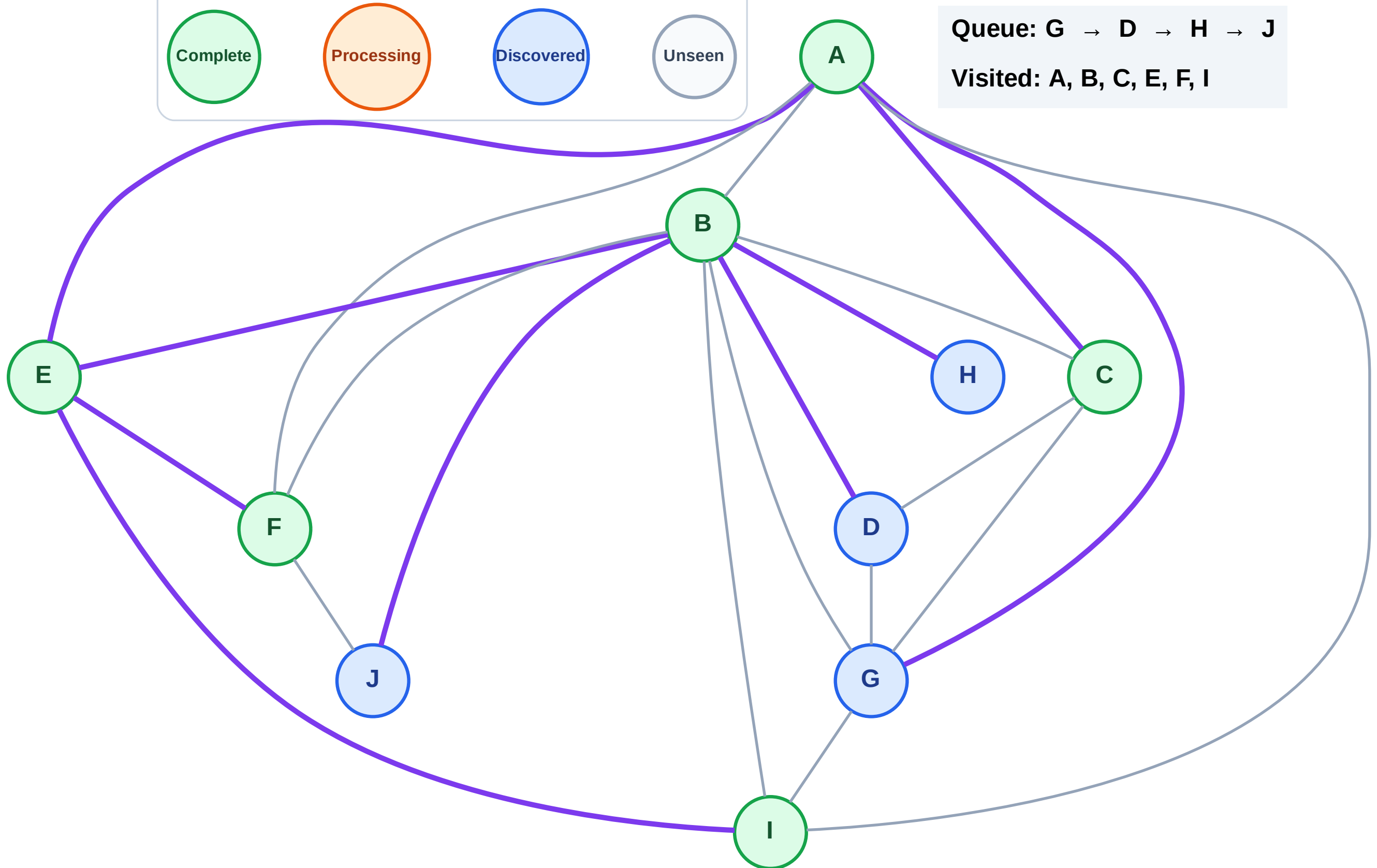
Step 13: No new neighbors from C

Legend



Queue: G → D → H → J

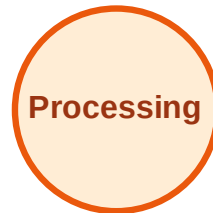
Visited: A, B, C, E, F, I



BFS Traversal

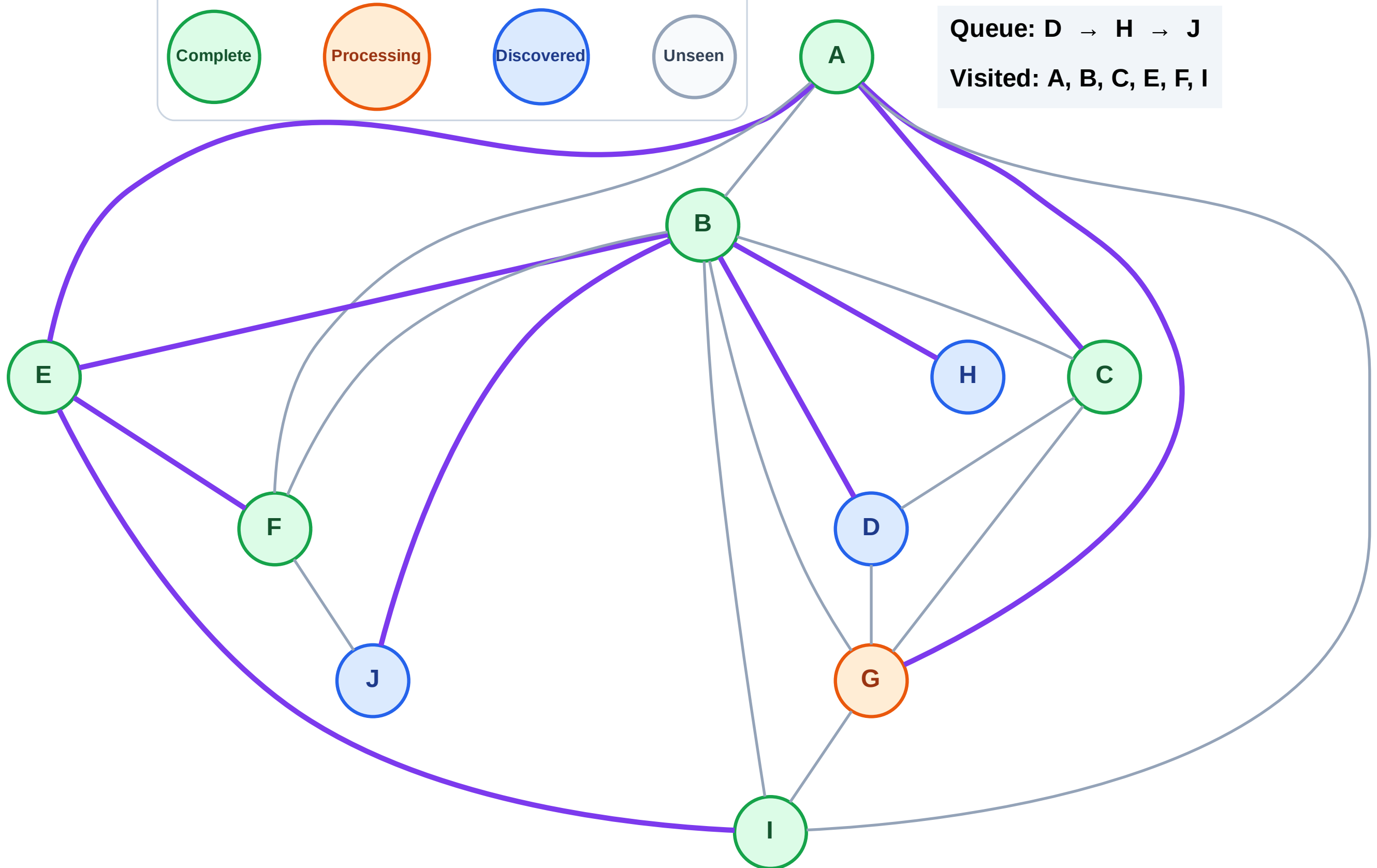
Step 14: Process node G

Legend



Queue: D → H → J

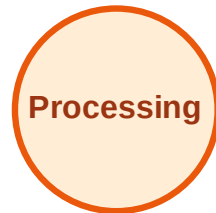
Visited: A, B, C, E, F, I



BFS Traversal

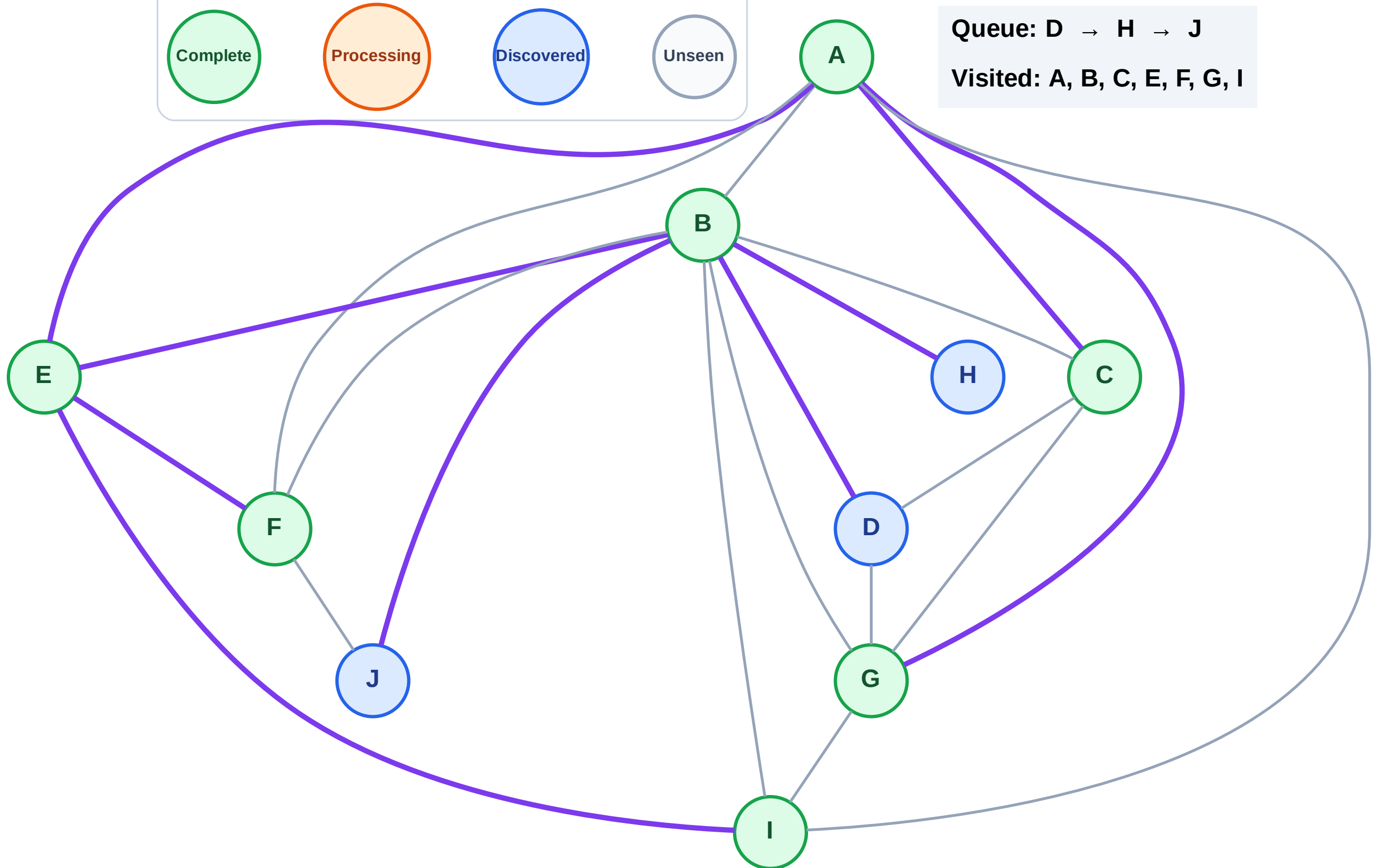
Step 15: No new neighbors from G

Legend



Queue: D → H → J

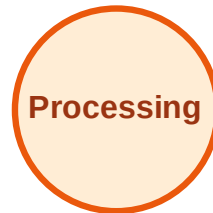
Visited: A, B, C, E, F, G, I



BFS Traversal

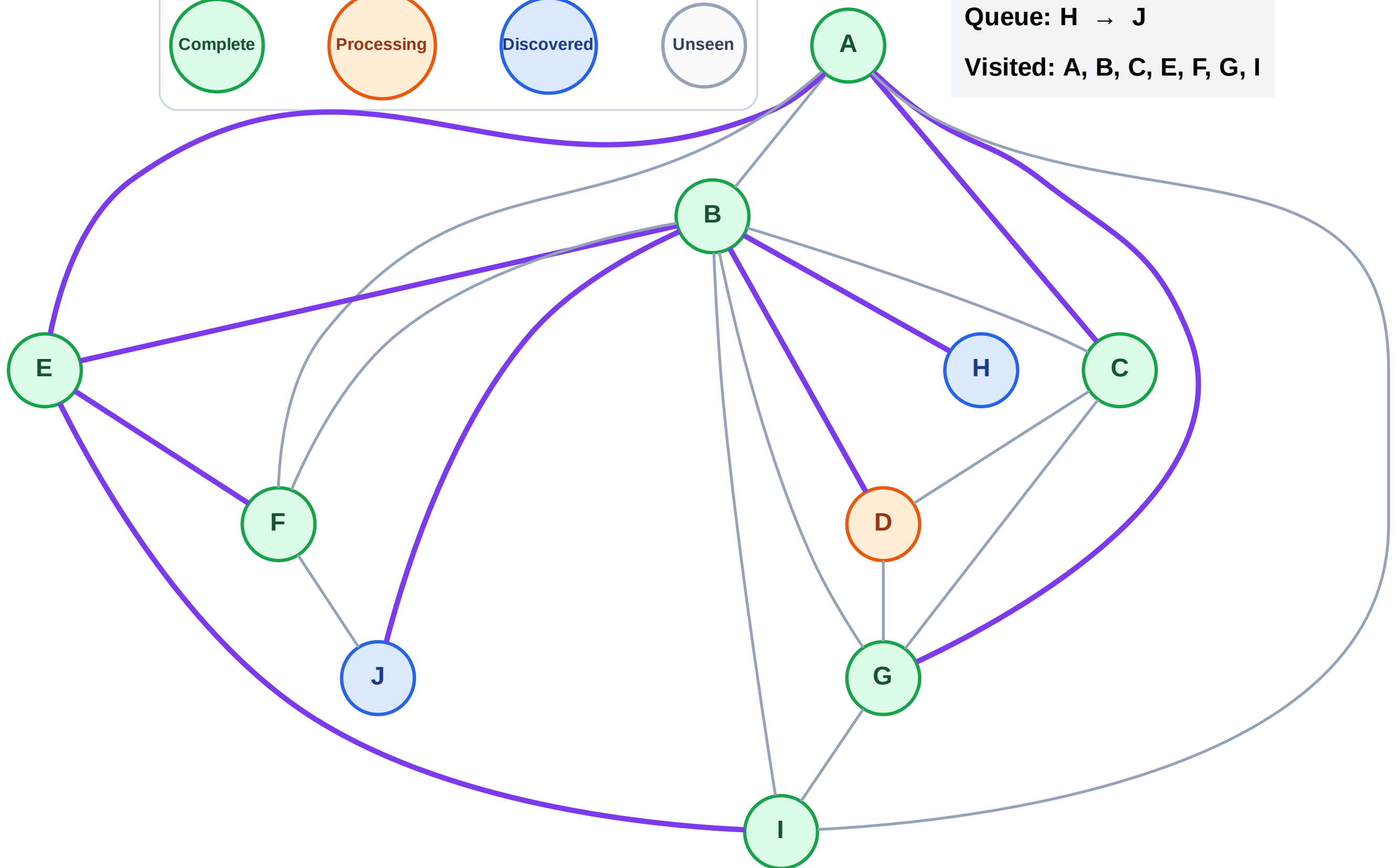
Step 16: Process node D

Legend



Queue: H → J

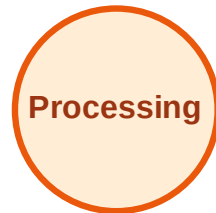
Visited: A, B, C, E, F, G, I



BFS Traversal

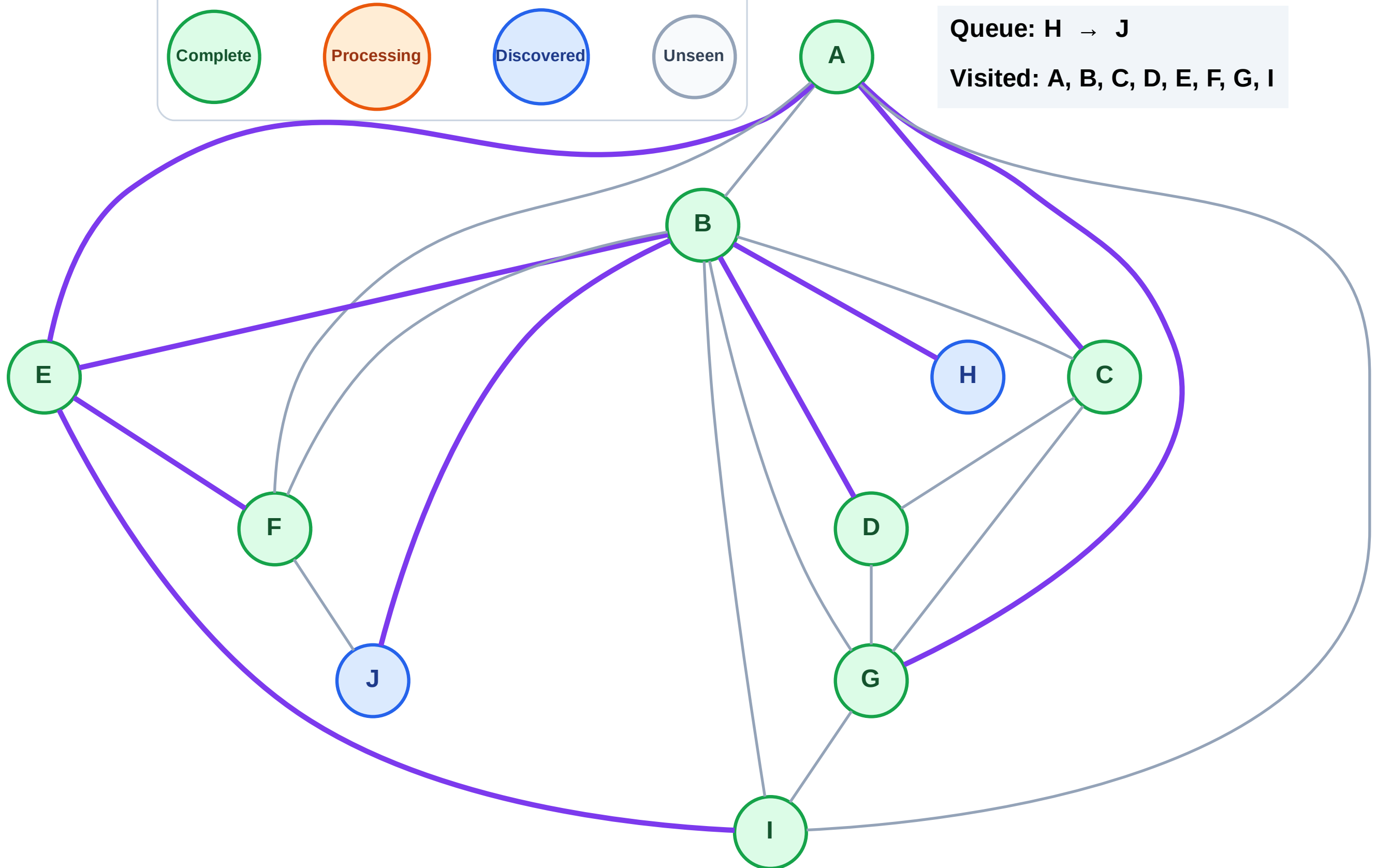
Step 17: No new neighbors from D

Legend



Queue: H → J

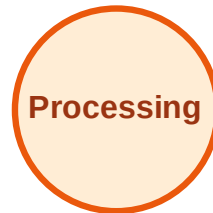
Visited: A, B, C, D, E, F, G, I



BFS Traversal

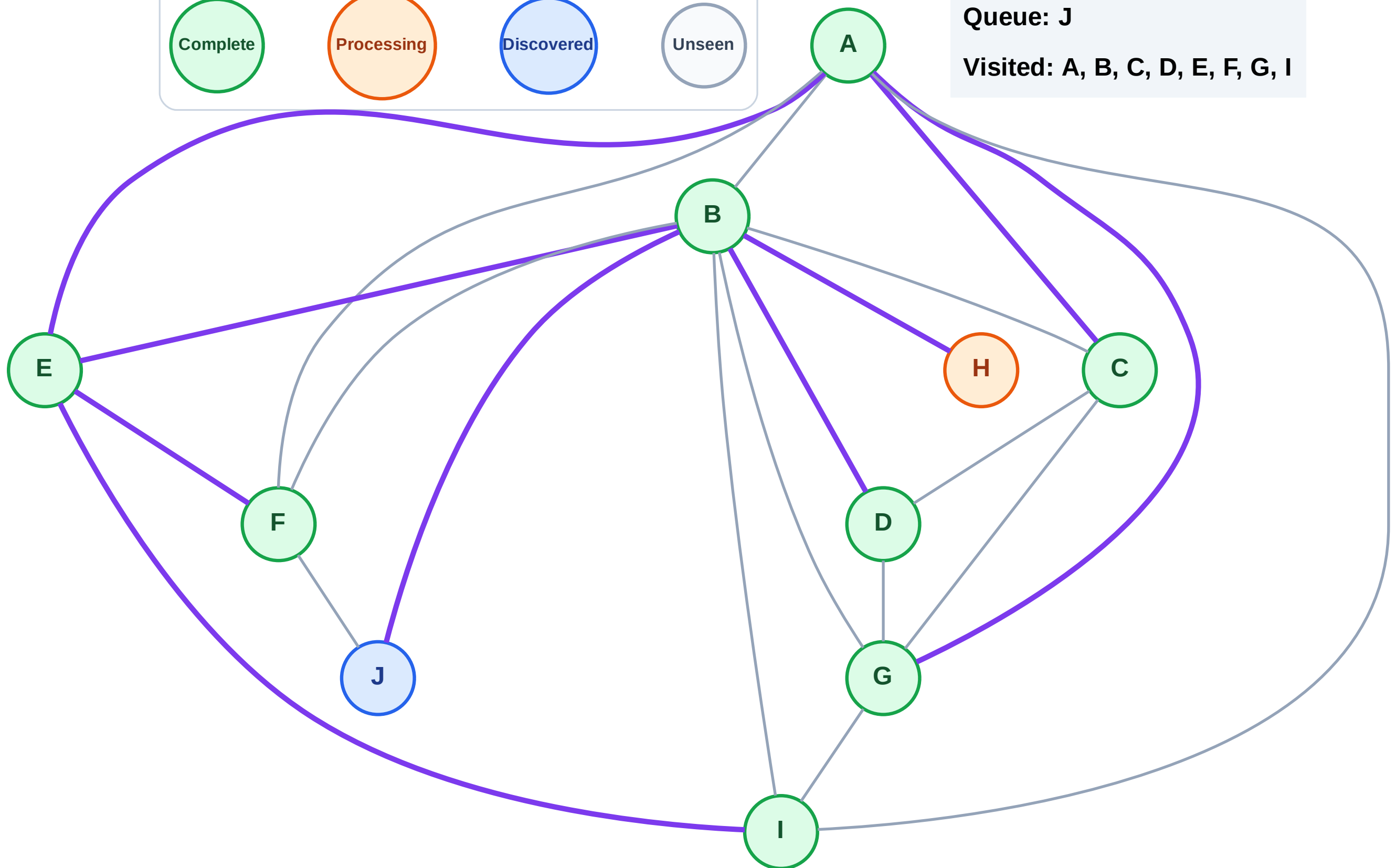
Step 18: Process node H

Legend



Queue: J

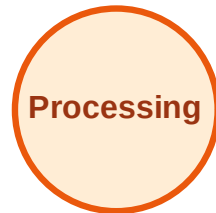
Visited: A, B, C, D, E, F, G, I



BFS Traversal

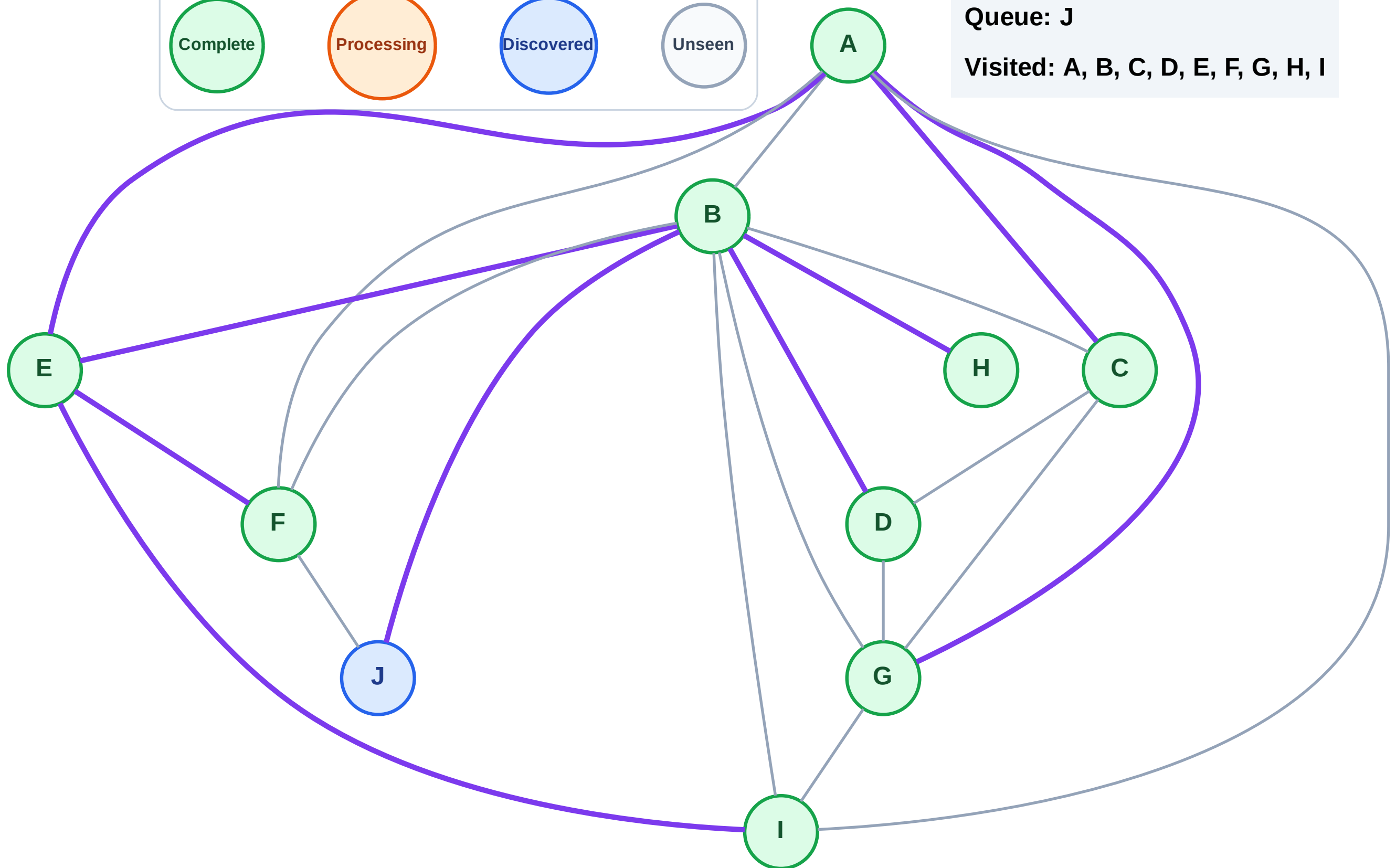
Step 19: No new neighbors from H

Legend



Queue: J

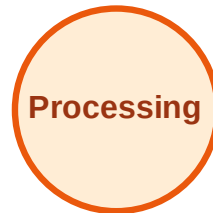
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

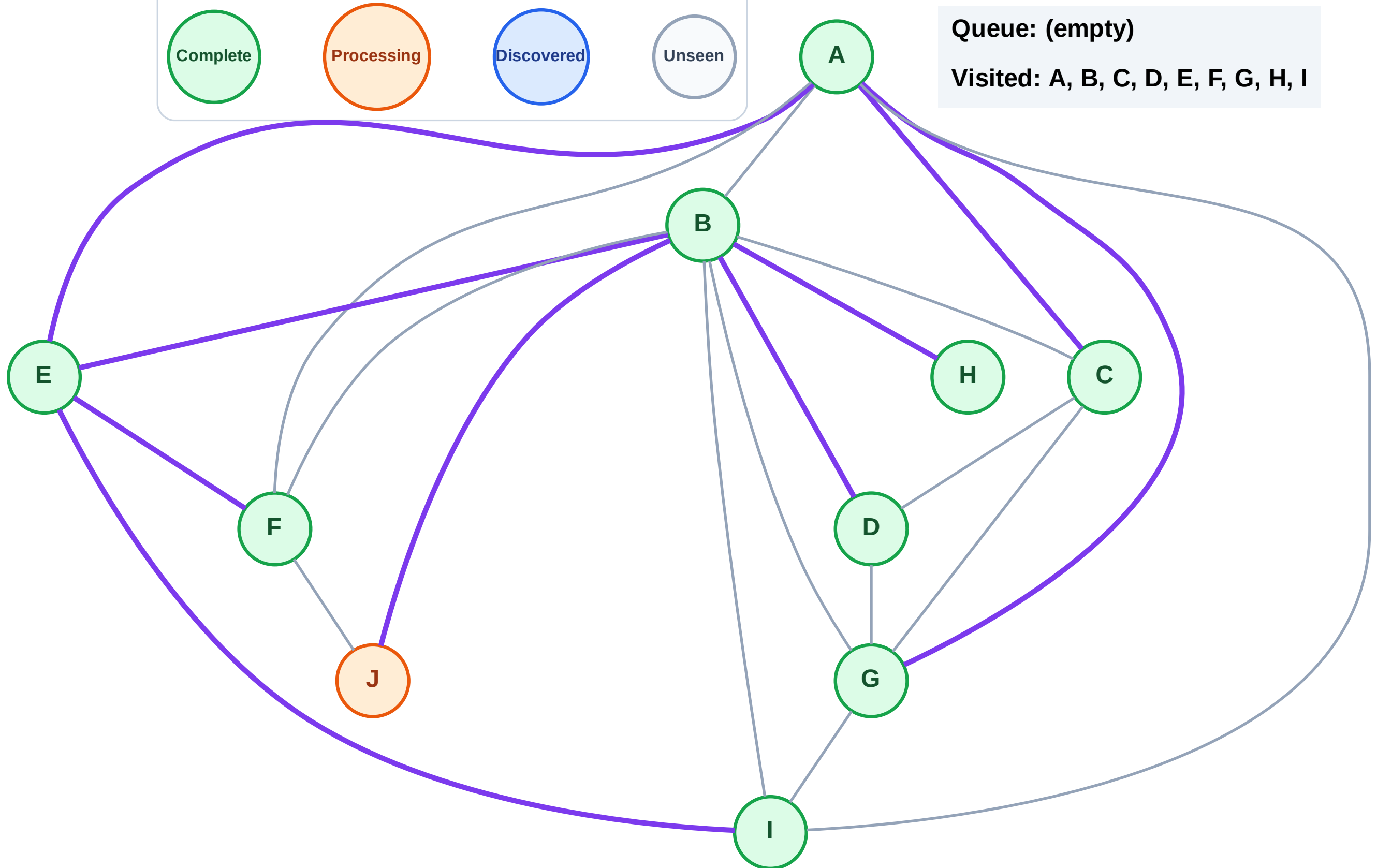
Step 20: Process node J

Legend



Queue: (empty)

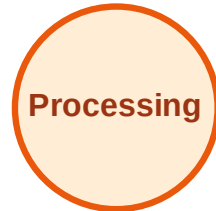
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

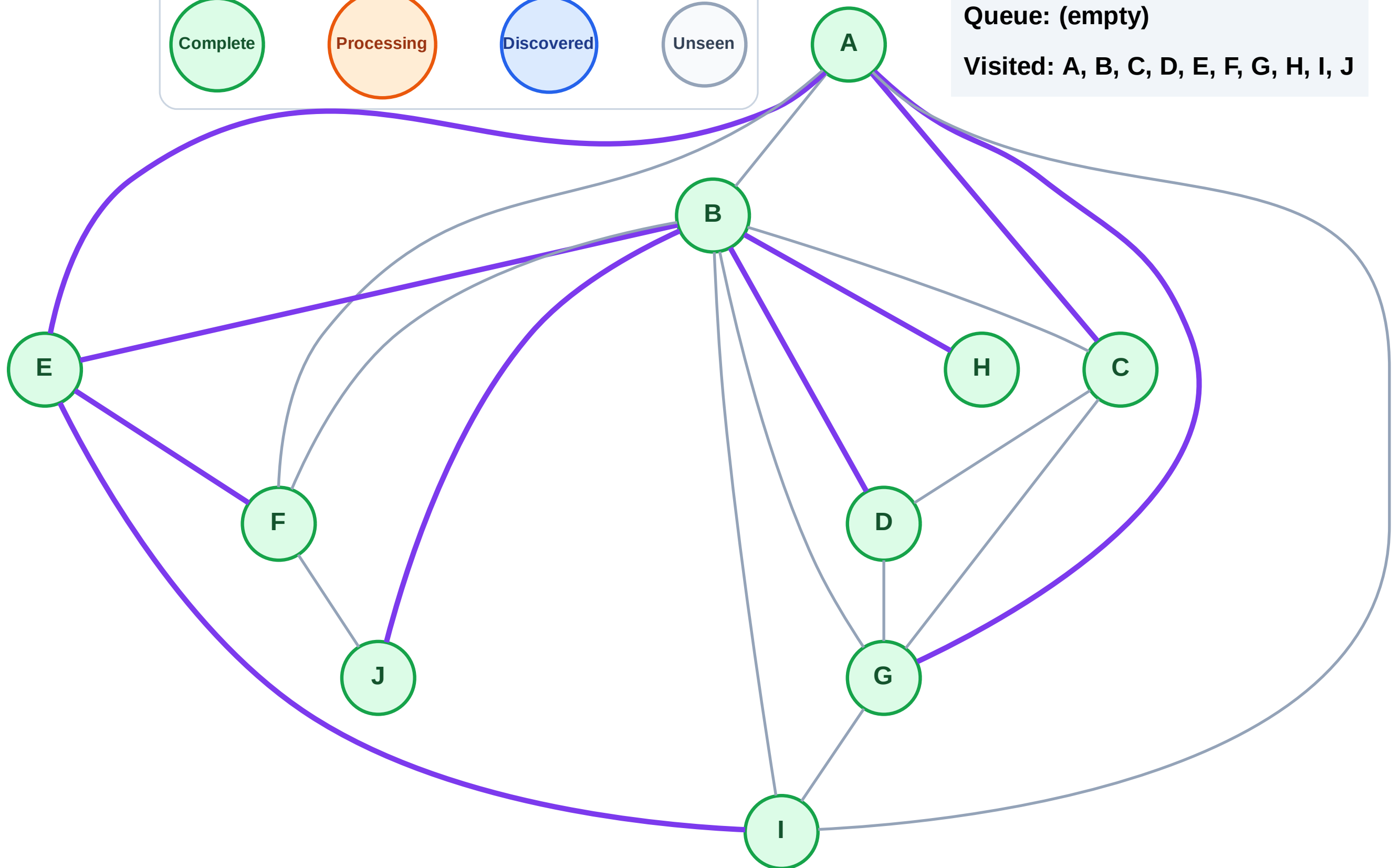
Step 21: No new neighbors from J

Legend



Queue: (empty)

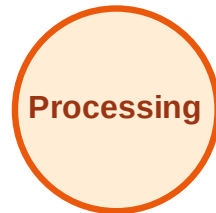
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

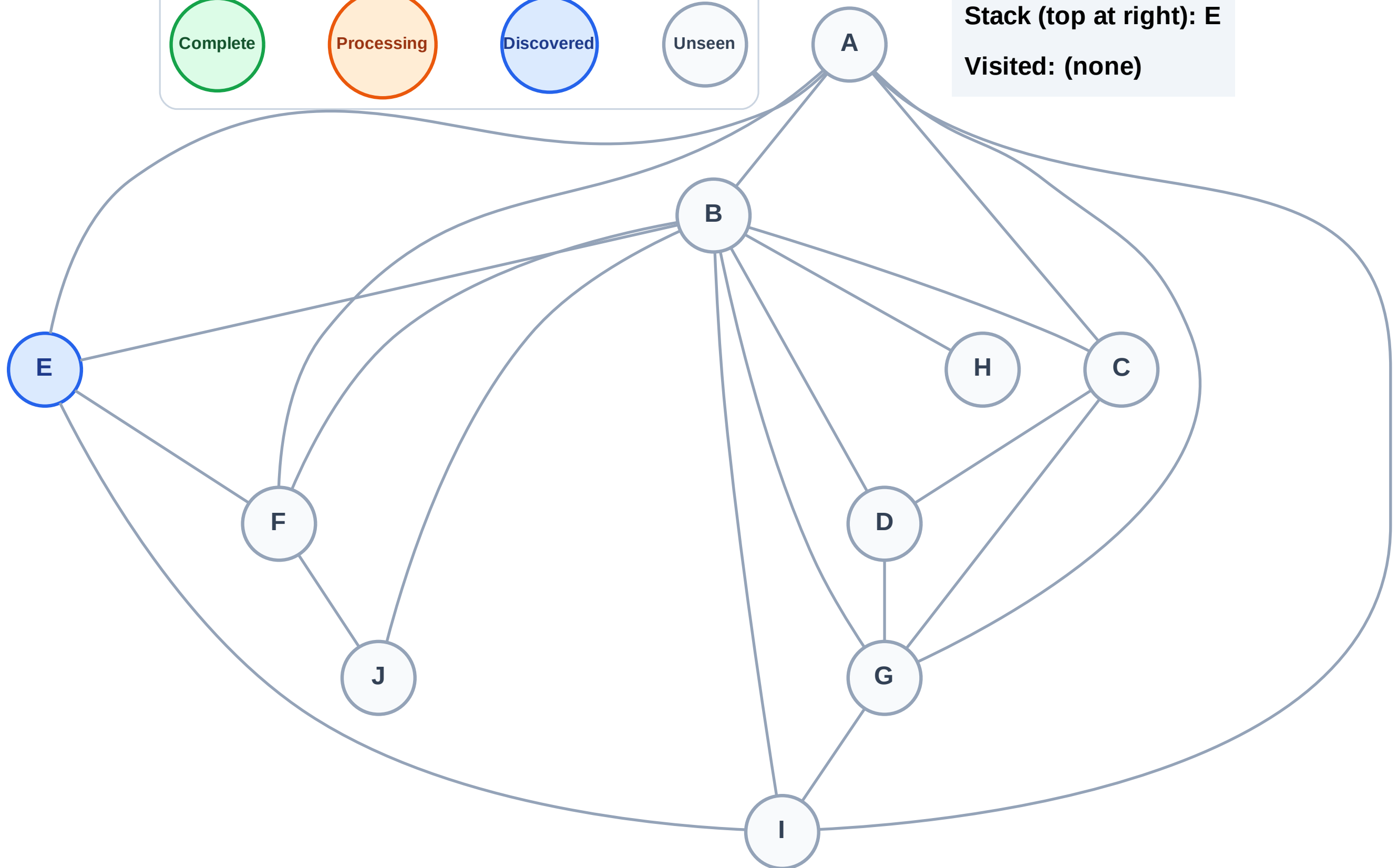
Step 1: Start at E

Legend



Stack (top at right): E

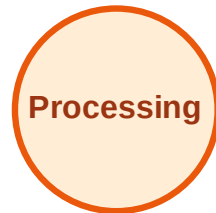
Visited: (none)



DFS Traversal

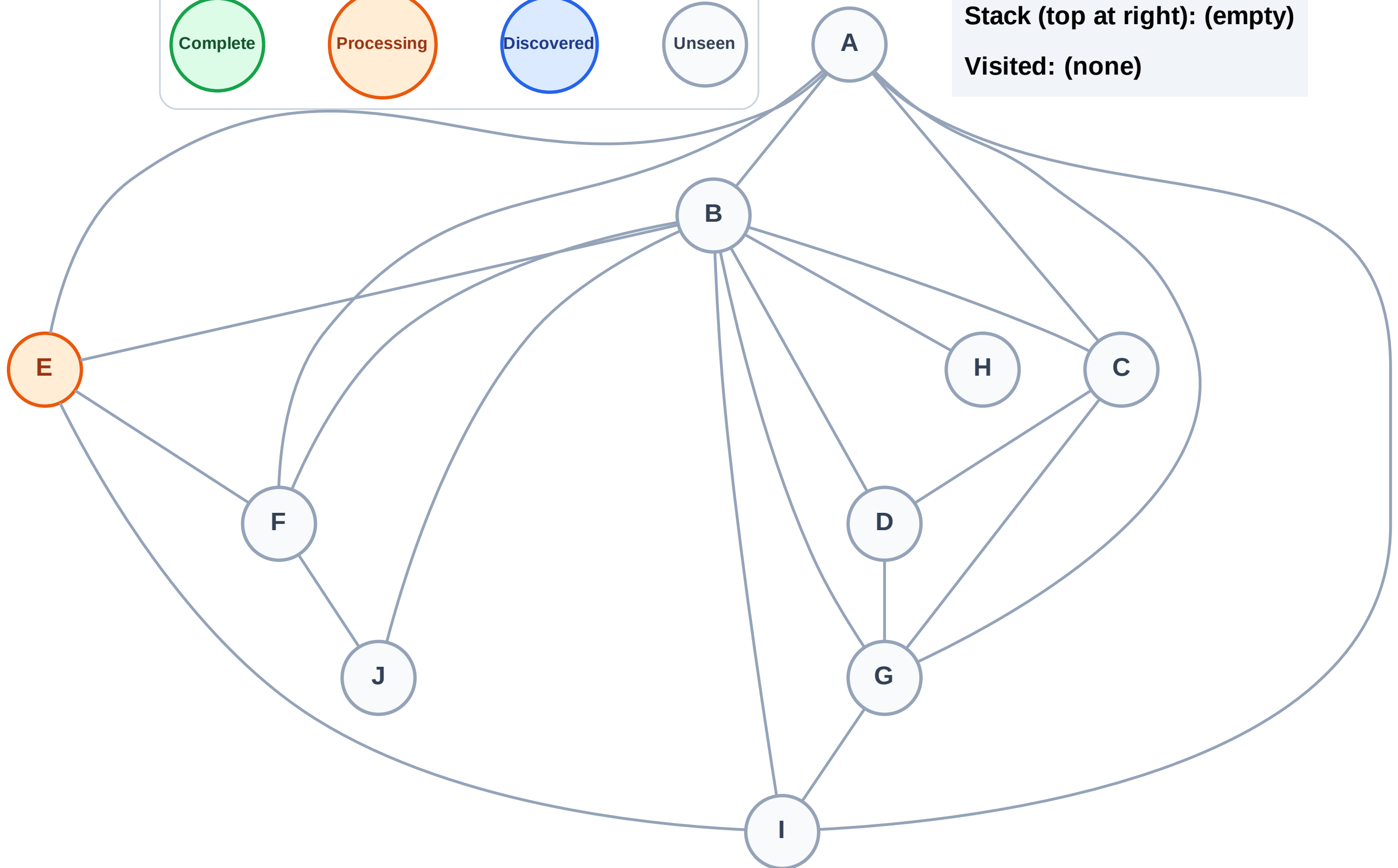
Step 2: Process node E

Legend



Stack (top at right): (empty)

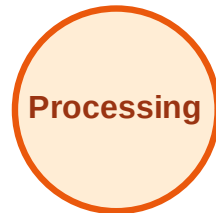
Visited: (none)



DFS Traversal

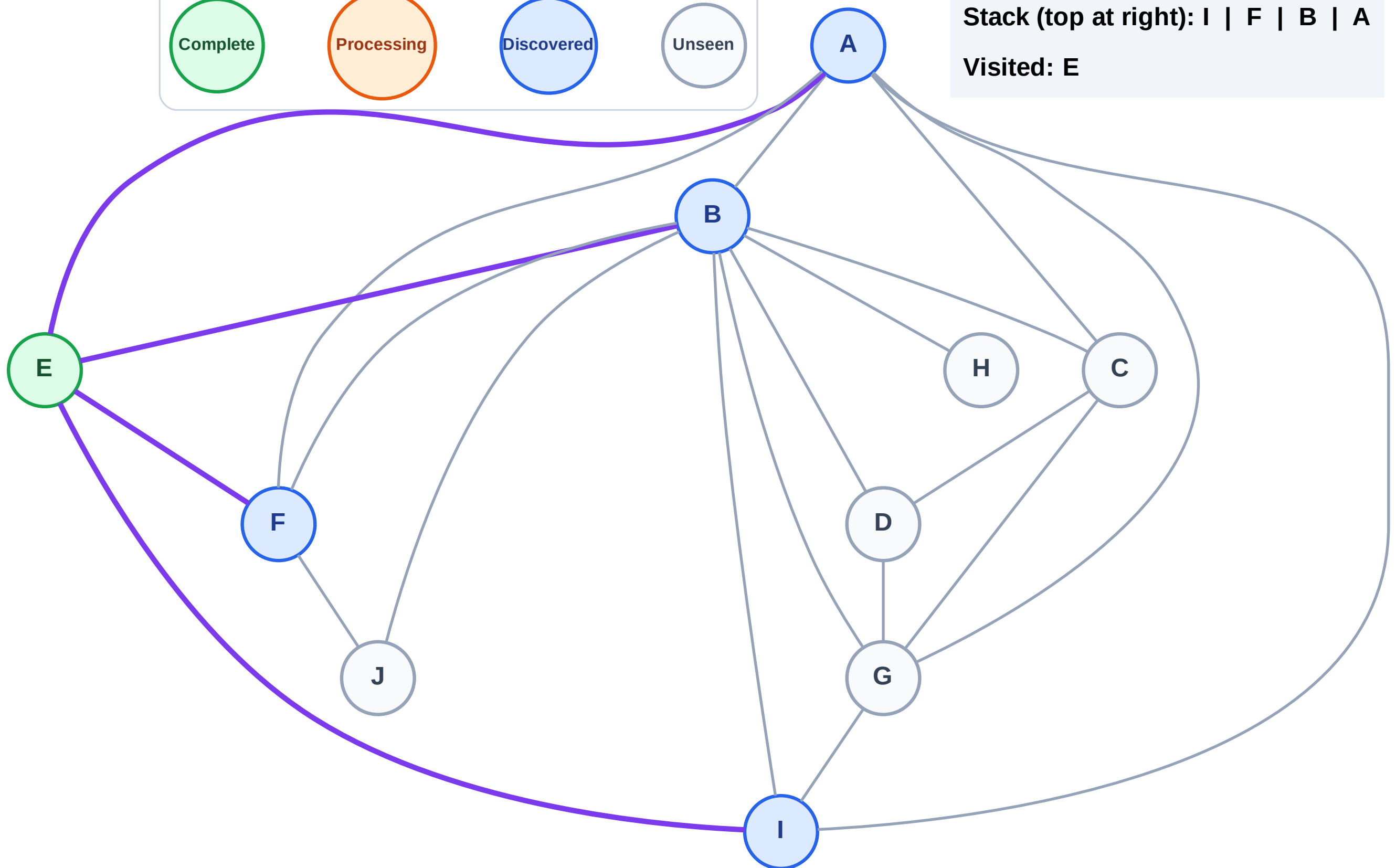
Step 3: Discover I, F, B, A from E

Legend



Stack (top at right): I | F | B | A

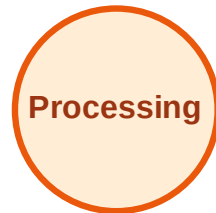
Visited: E



DFS Traversal

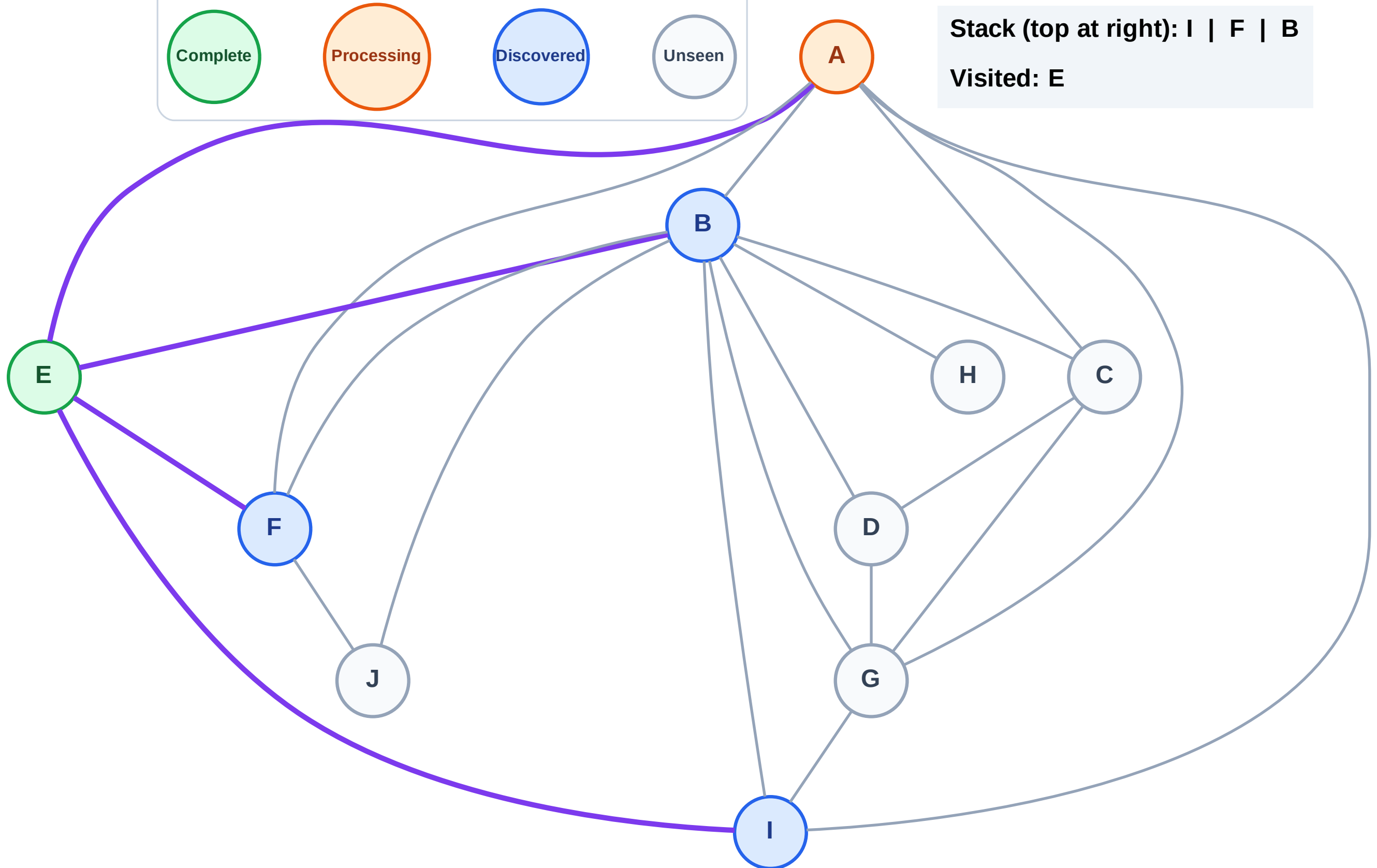
Step 4: Process node A

Legend



Stack (top at right): I | F | B

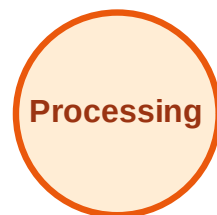
Visited: E



DFS Traversal

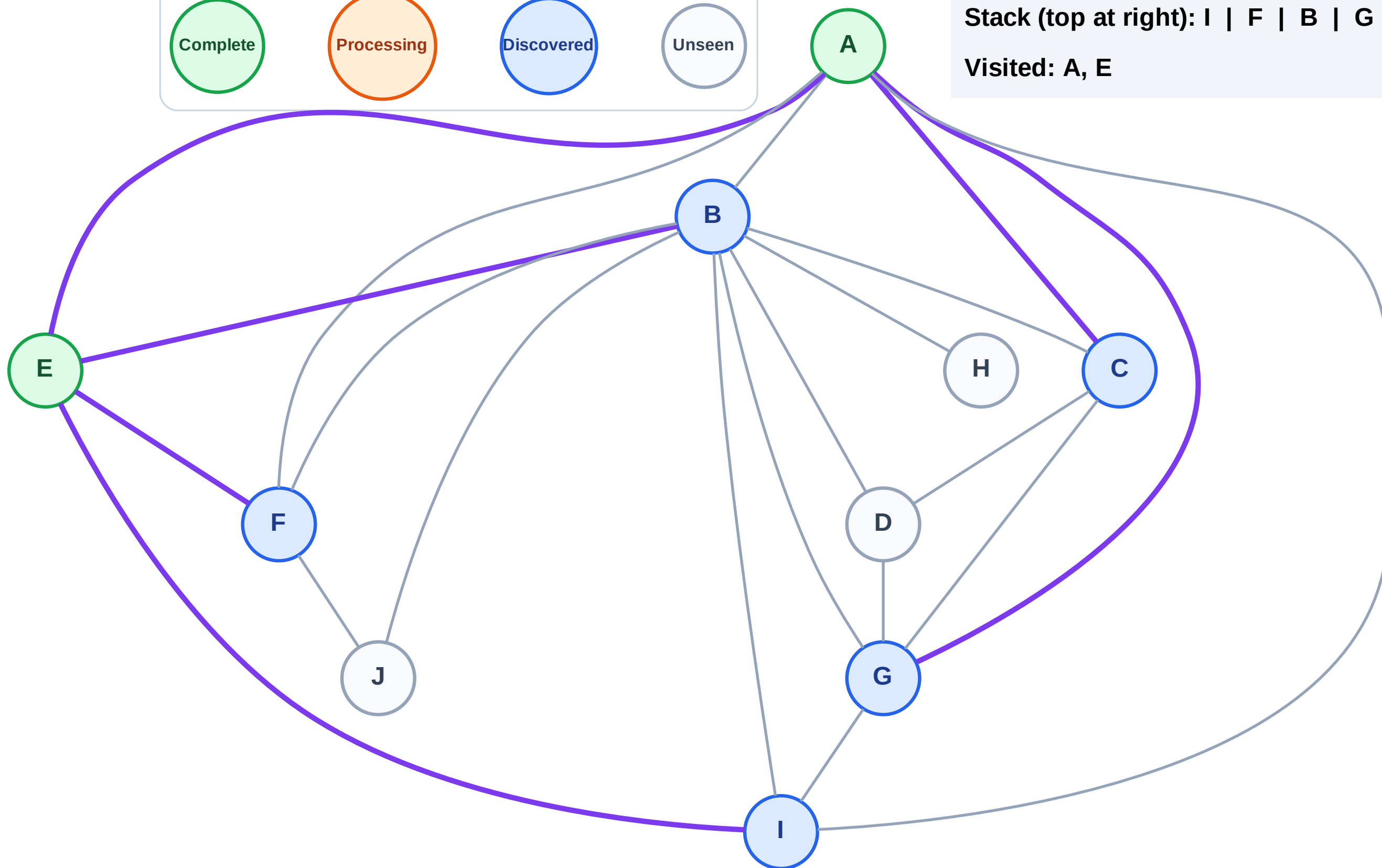
Step 5: Discover G, C from A

Legend



Stack (top at right): I | F | B | G | C

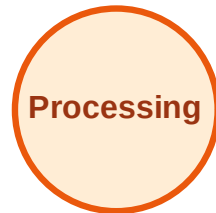
Visited: A, E



DFS Traversal

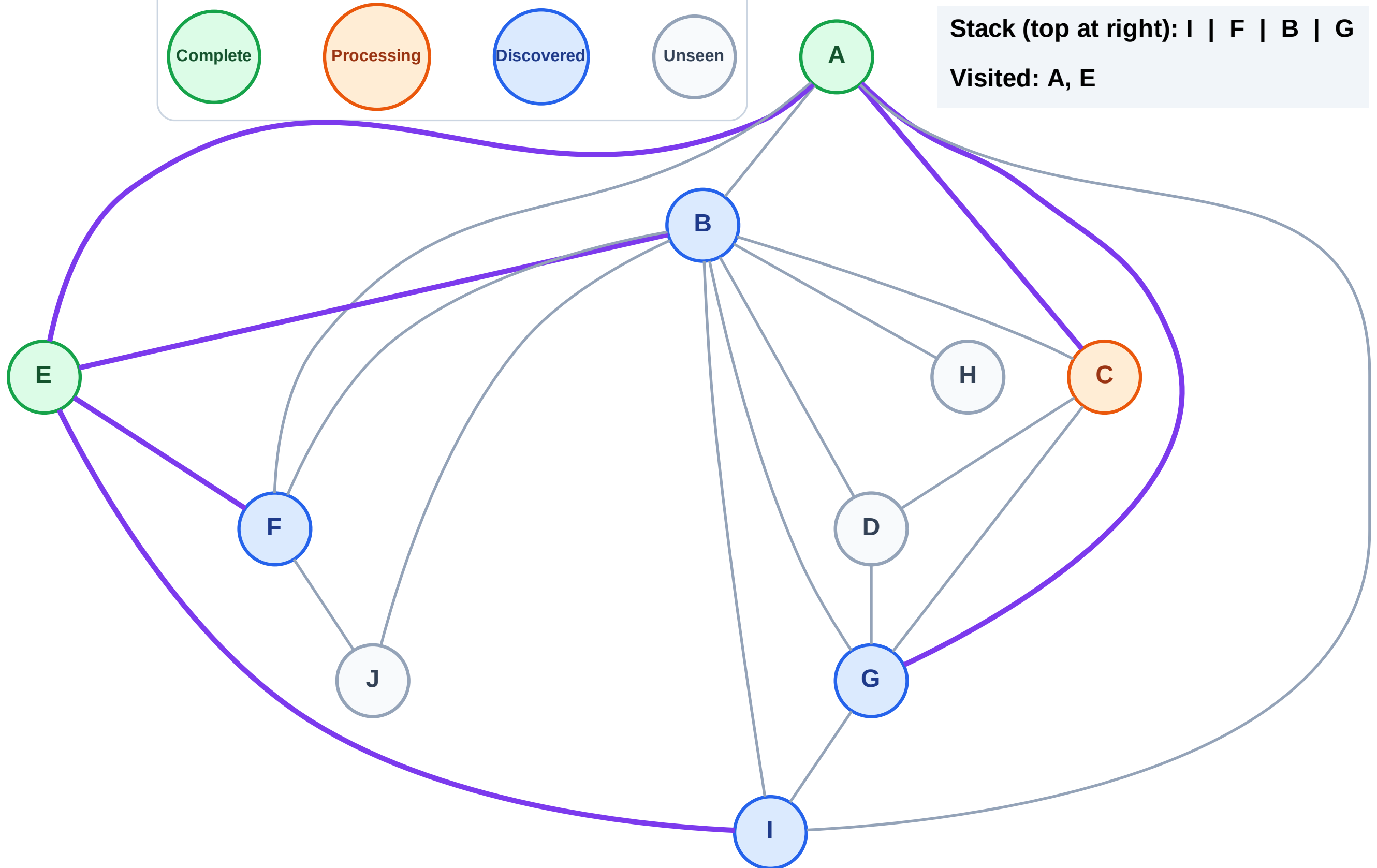
Step 6: Process node C

Legend



Stack (top at right): I | F | B | G

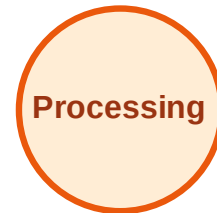
Visited: A, E



DFS Traversal

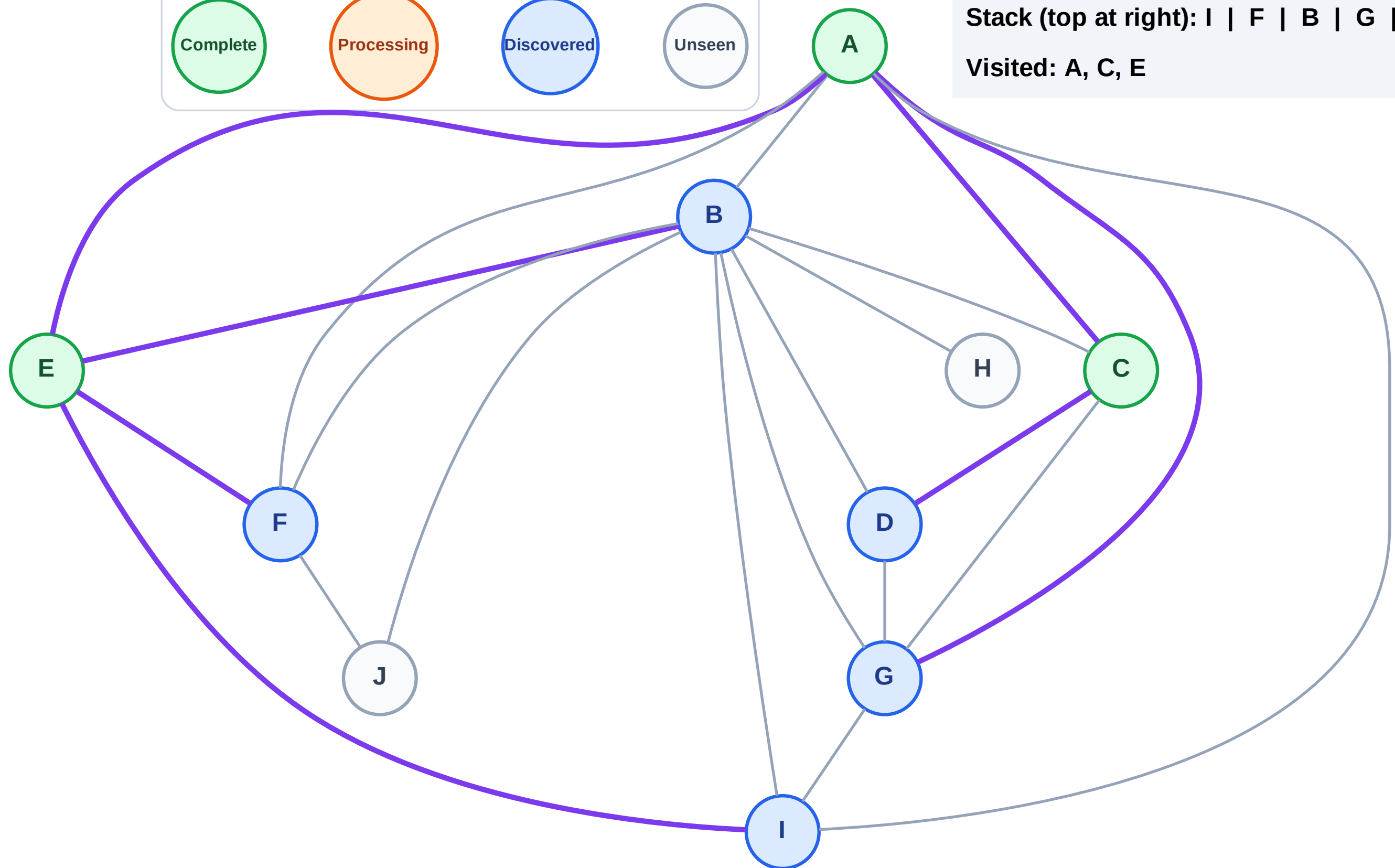
Step 7: Discover D from C

Legend



Stack (top at right): I | F | B | G | D

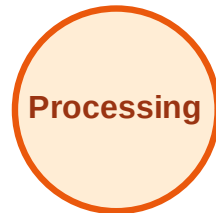
Visited: A, C, E



DFS Traversal

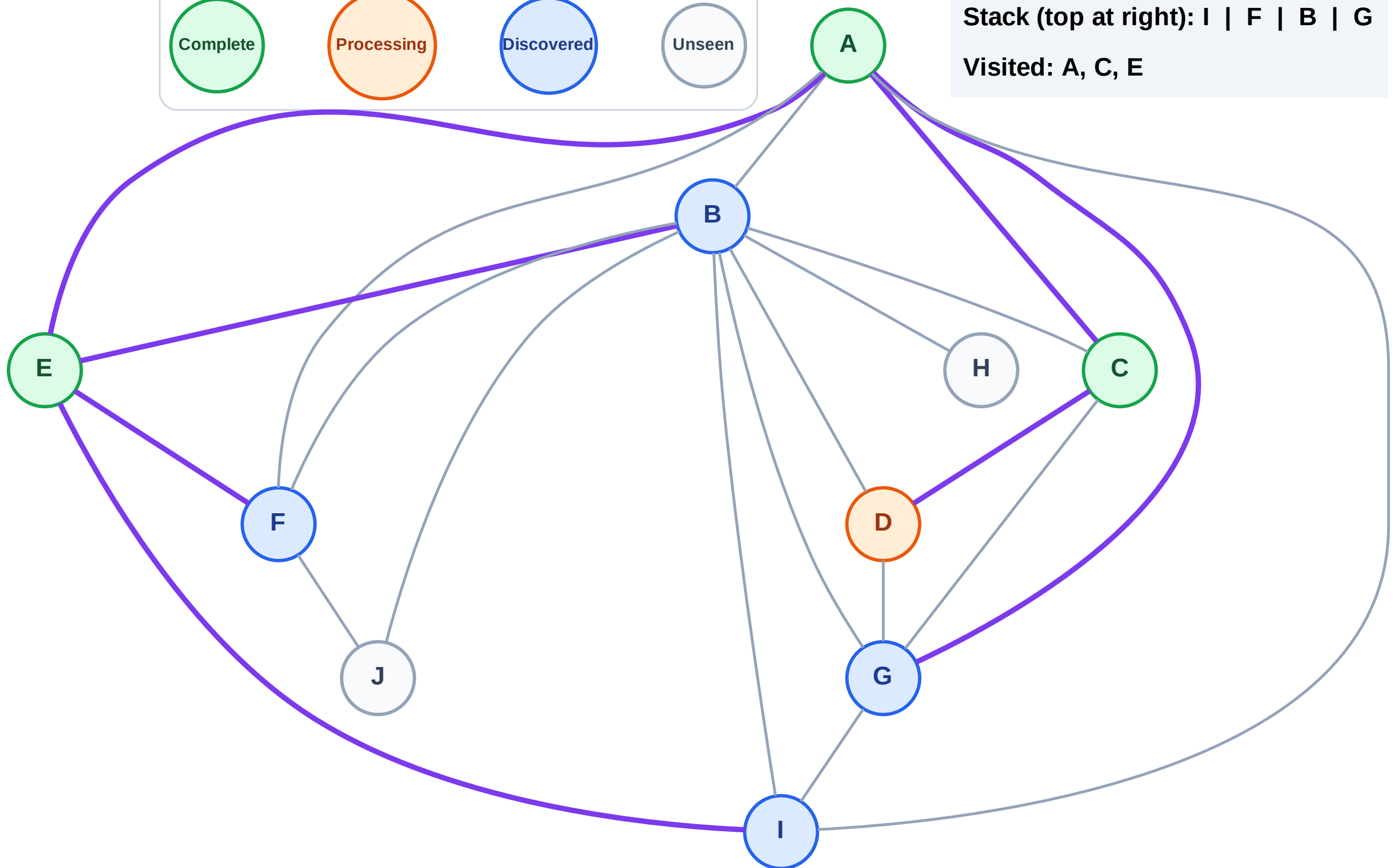
Step 8: Process node D

Legend



Stack (top at right): I | F | B | G

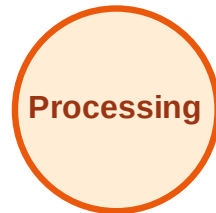
Visited: A, C, E



DFS Traversal

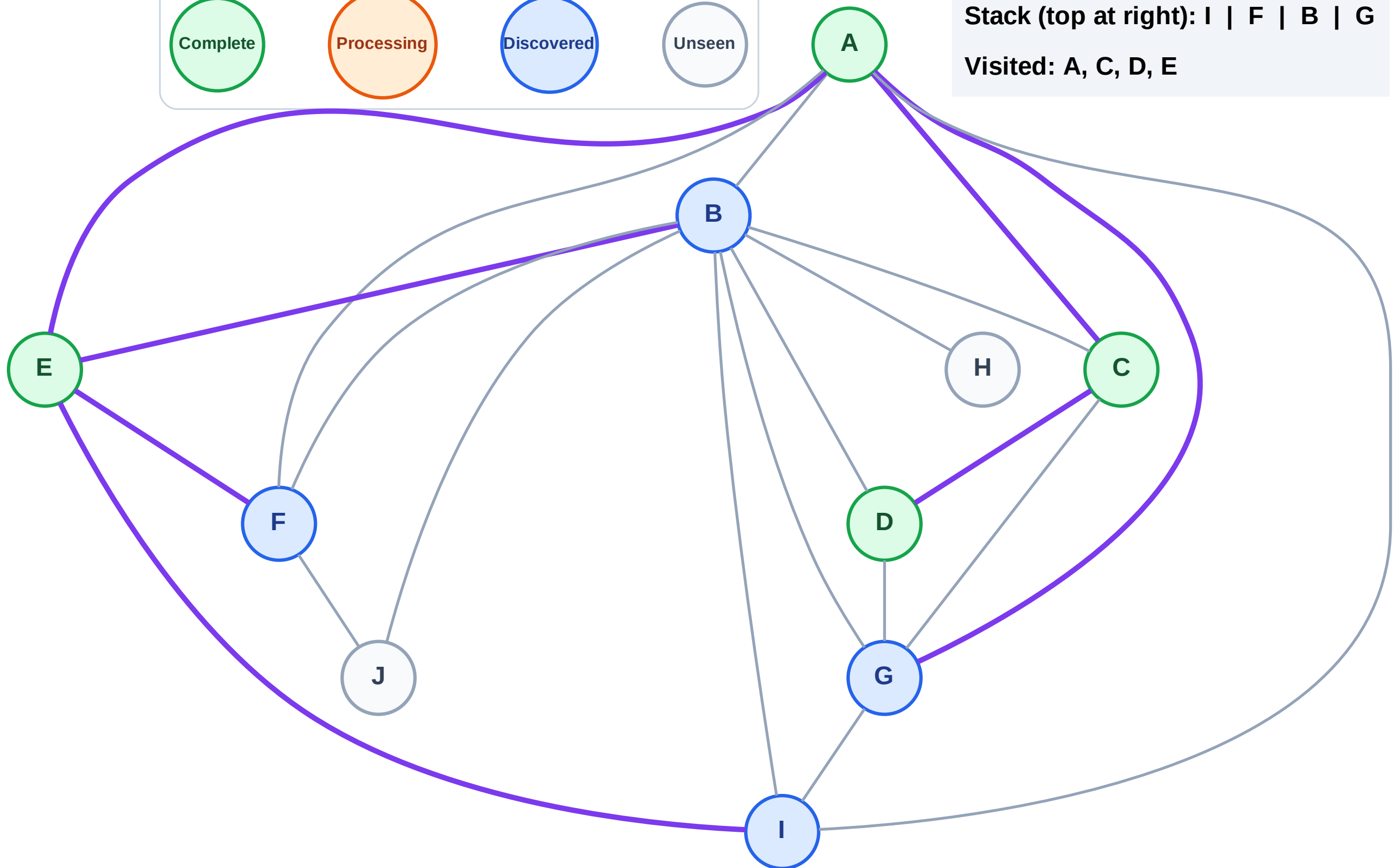
Step 9: No new neighbors from D

Legend



Stack (top at right): I | F | B | G

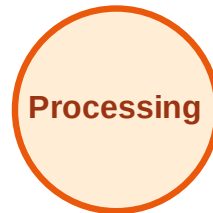
Visited: A, C, D, E



DFS Traversal

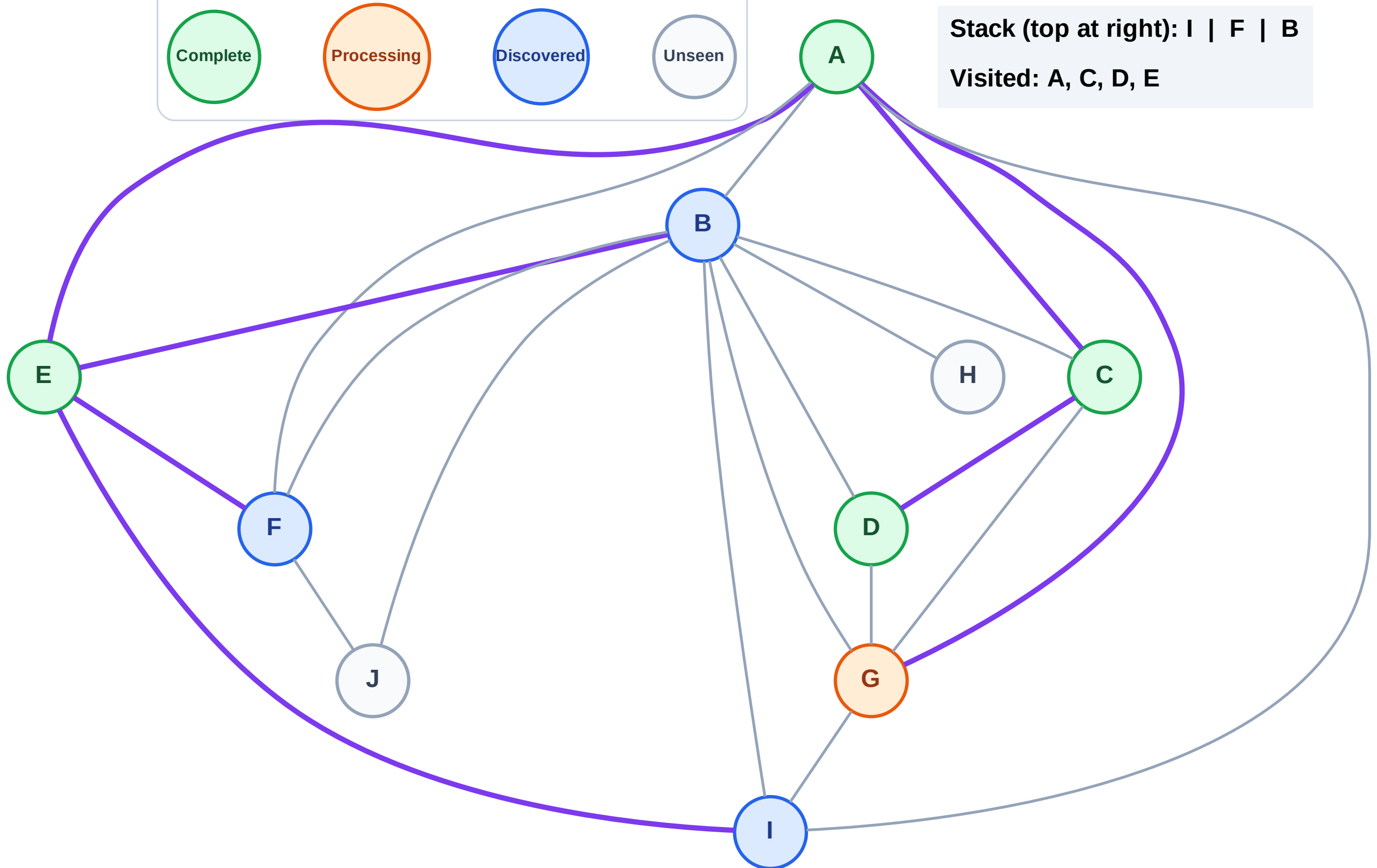
Step 10: Process node G

Legend



Stack (top at right): I | F | B

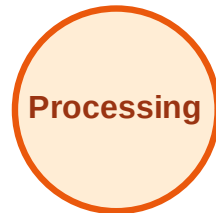
Visited: A, C, D, E



DFS Traversal

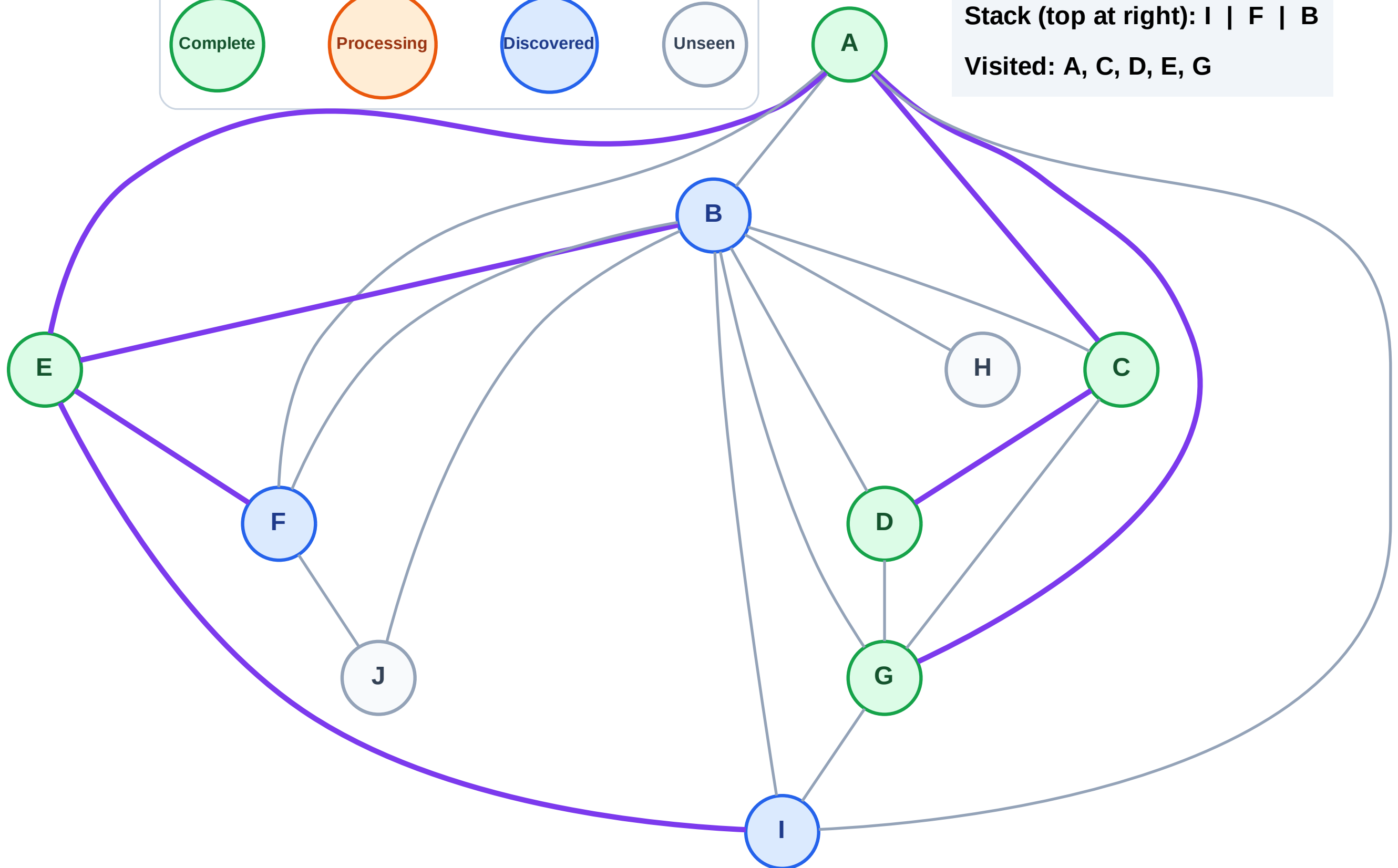
Step 11: No new neighbors from G

Legend



Stack (top at right): I | F | B

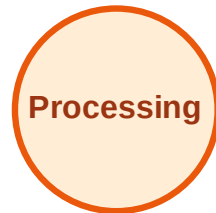
Visited: A, C, D, E, G



DFS Traversal

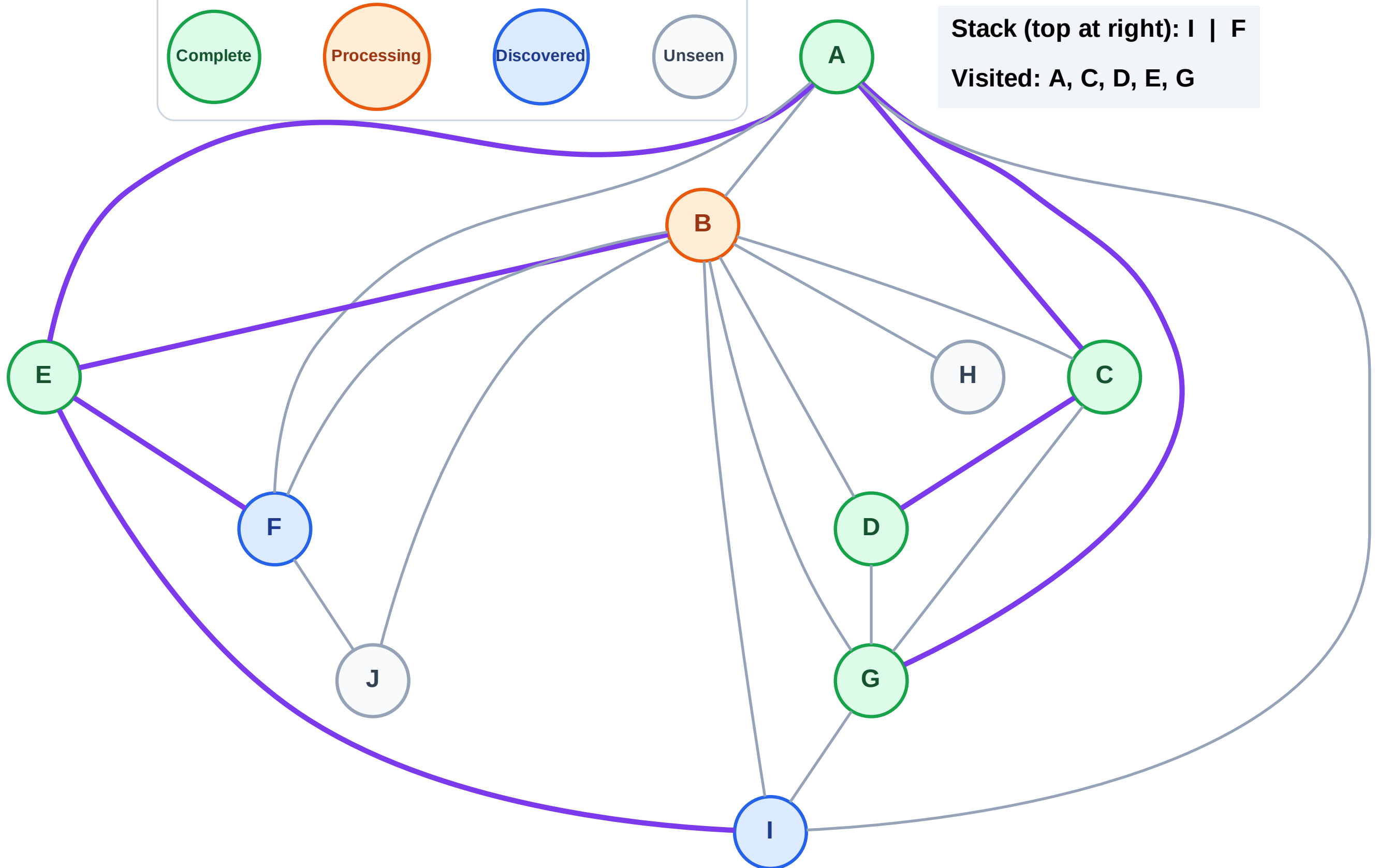
Step 12: Process node B

Legend



Stack (top at right): I | F

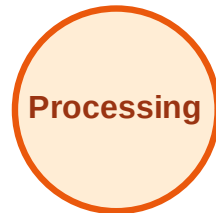
Visited: A, C, D, E, G



DFS Traversal

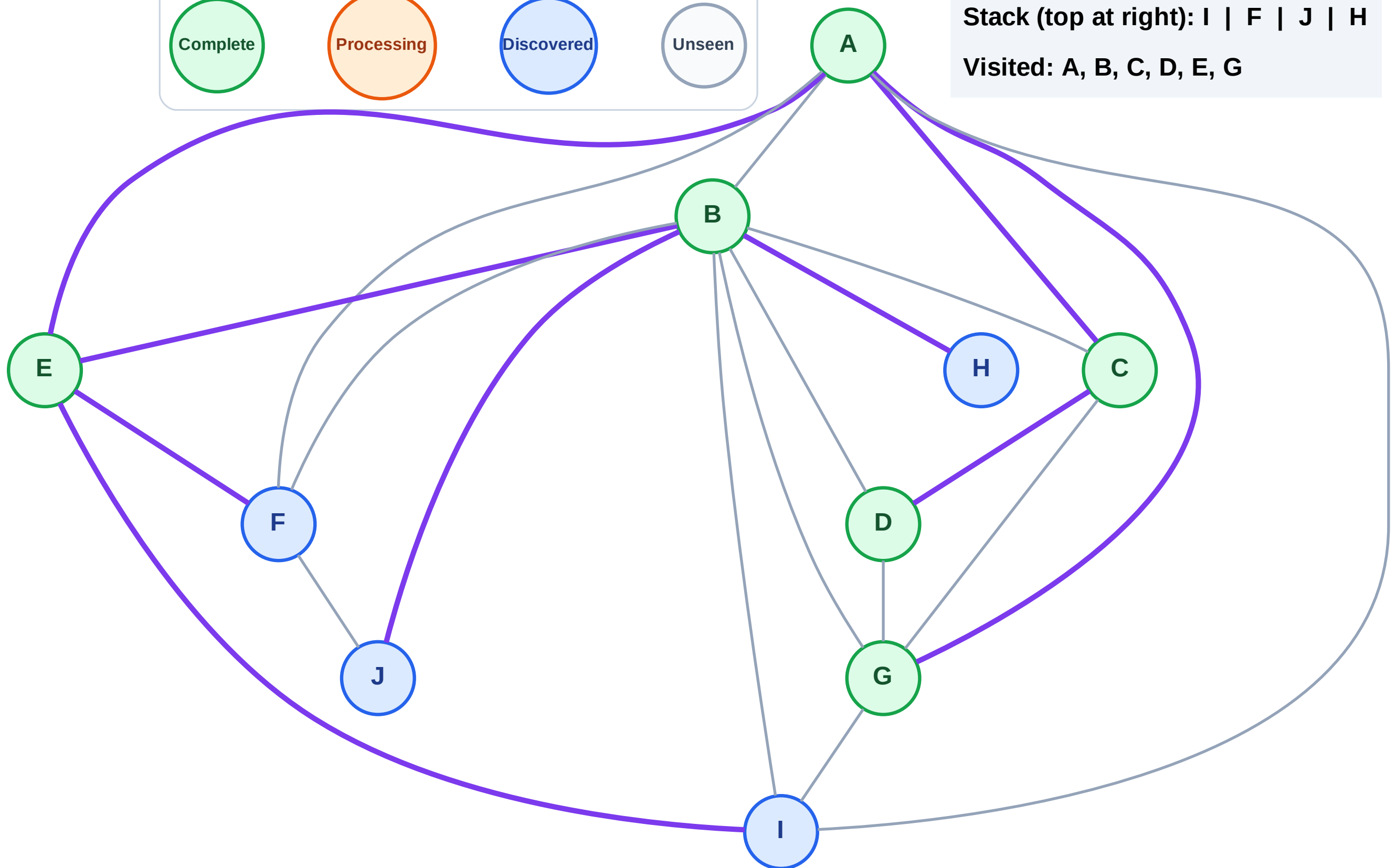
Step 13: Discover J, H from B

Legend



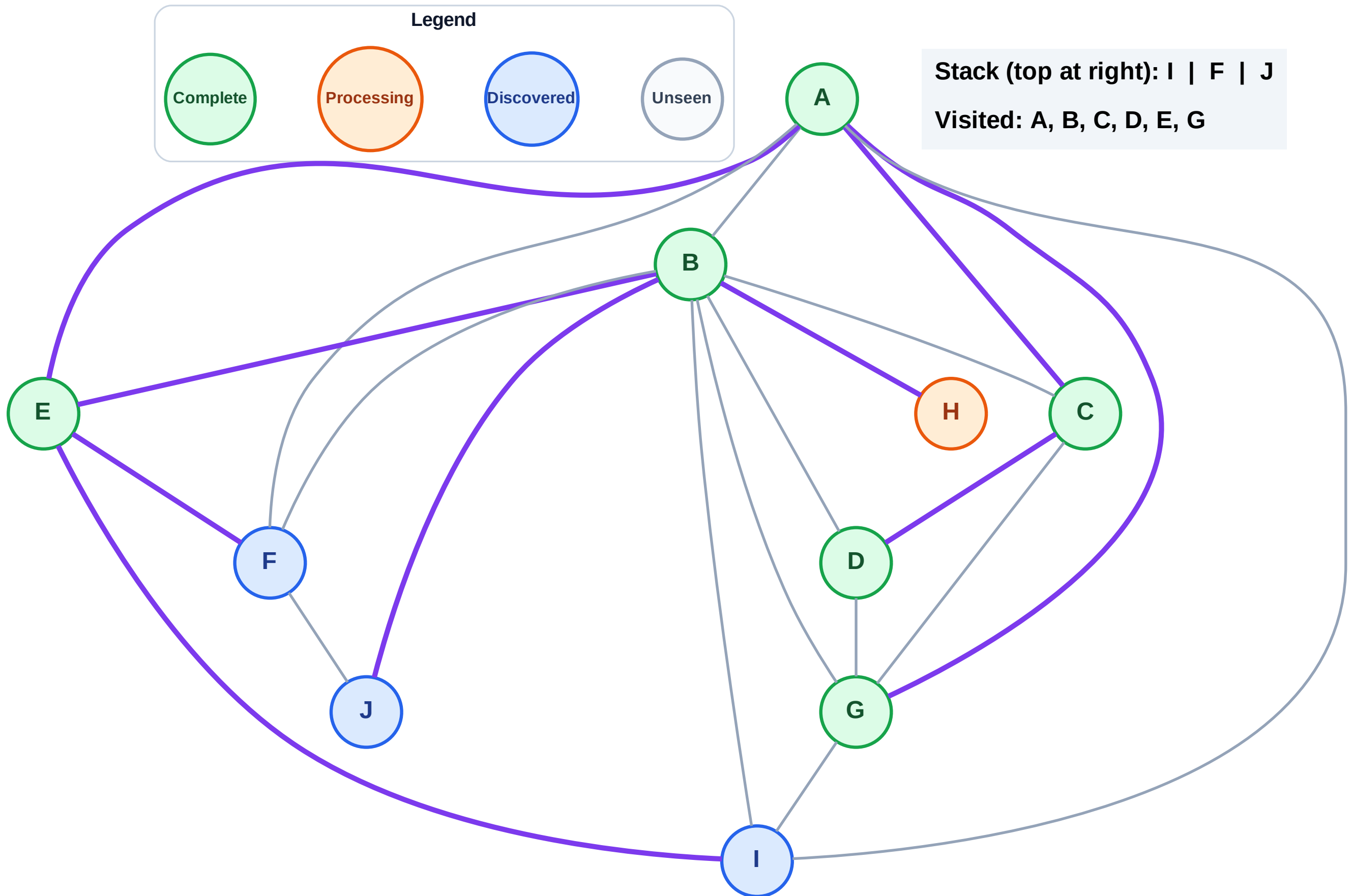
Stack (top at right): I | F | J | H

Visited: A, B, C, D, E, G



Step 14: Process node H

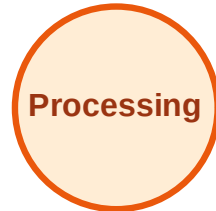
Step 14: Process node H



DFS Traversal

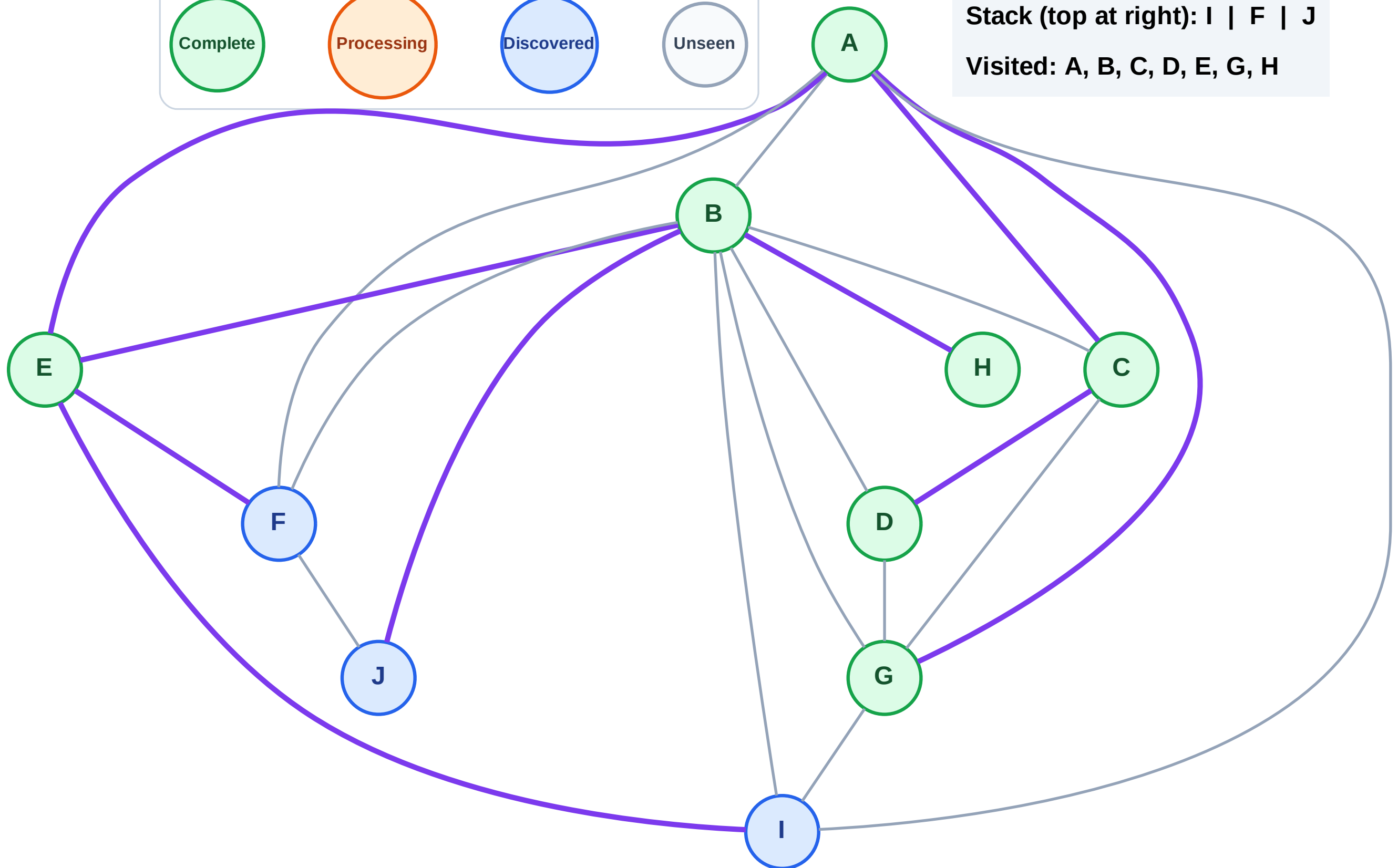
Step 15: No new neighbors from H

Legend



Stack (top at right): I | F | J

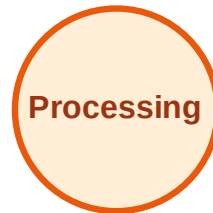
Visited: A, B, C, D, E, G, H



DFS Traversal

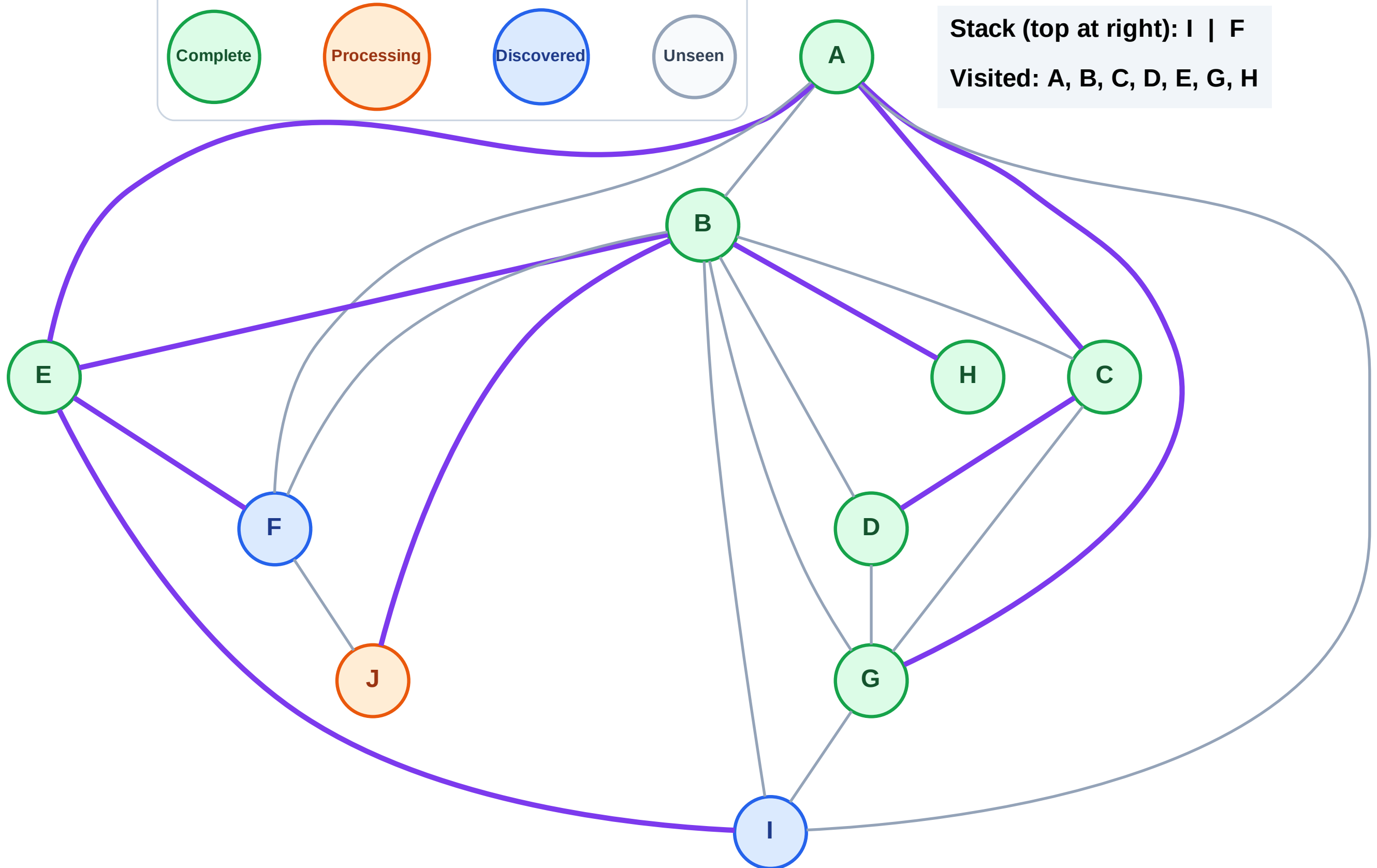
Step 16: Process node J

Legend



Stack (top at right): I | F

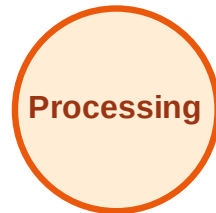
Visited: A, B, C, D, E, G, H



DFS Traversal

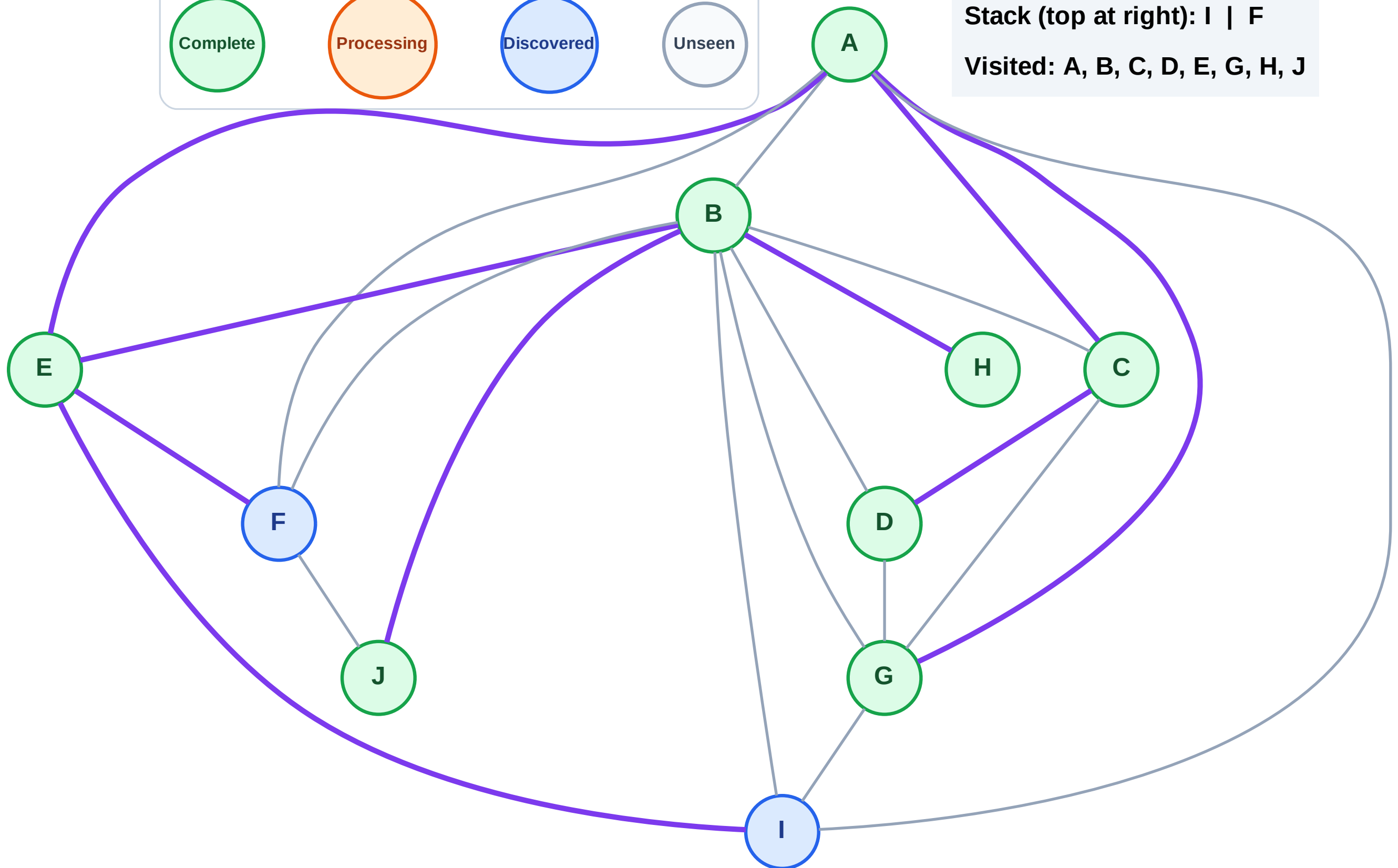
Step 17: No new neighbors from J

Legend



Stack (top at right): I | F

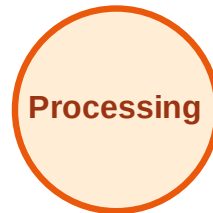
Visited: A, B, C, D, E, G, H, J



DFS Traversal

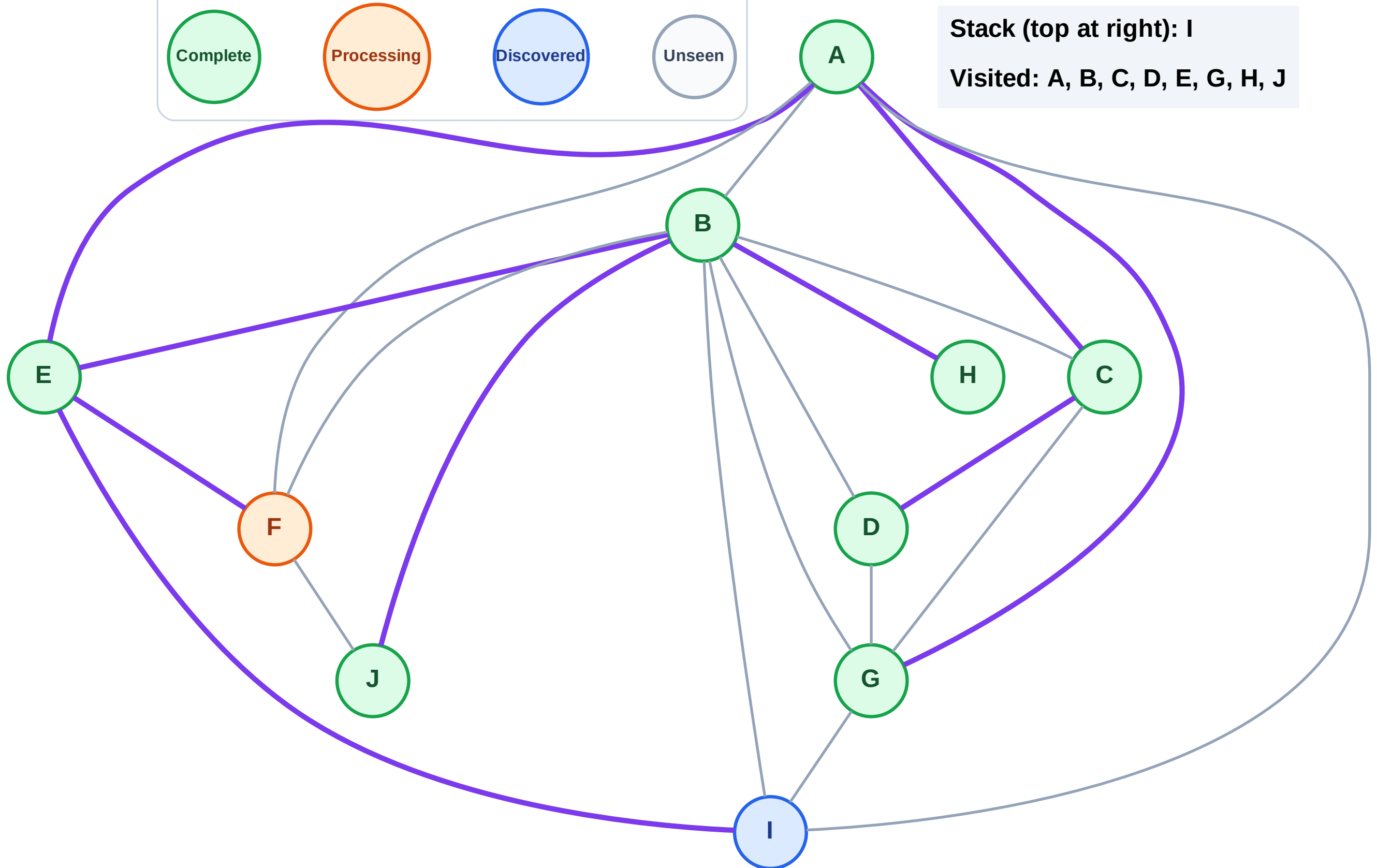
Step 18: Process node F

Legend



Stack (top at right): I

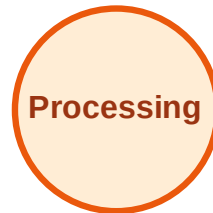
Visited: A, B, C, D, E, G, H, J



DFS Traversal

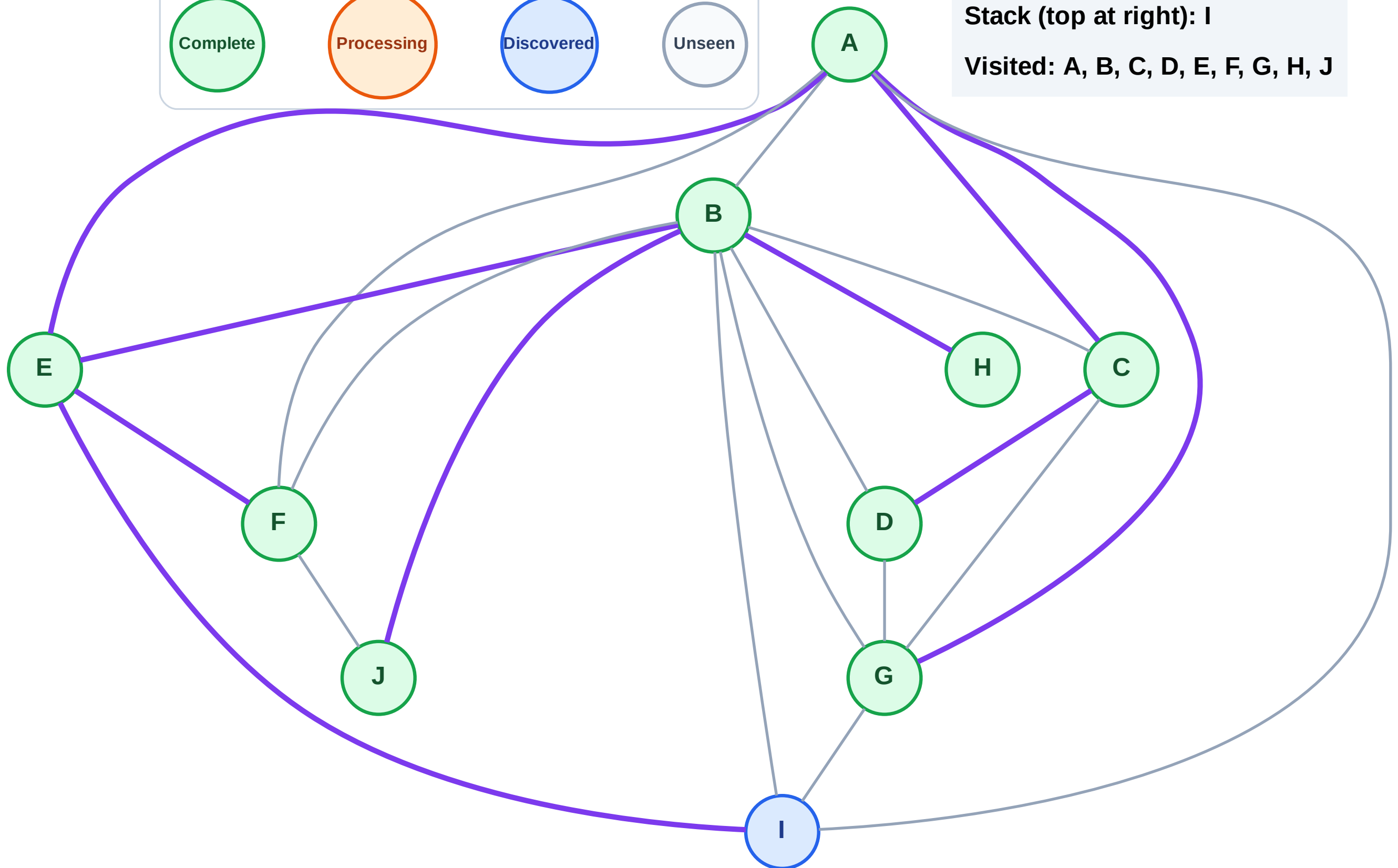
Step 19: No new neighbors from F

Legend



Stack (top at right): I

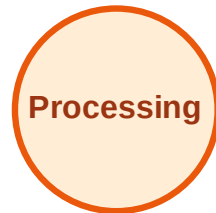
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

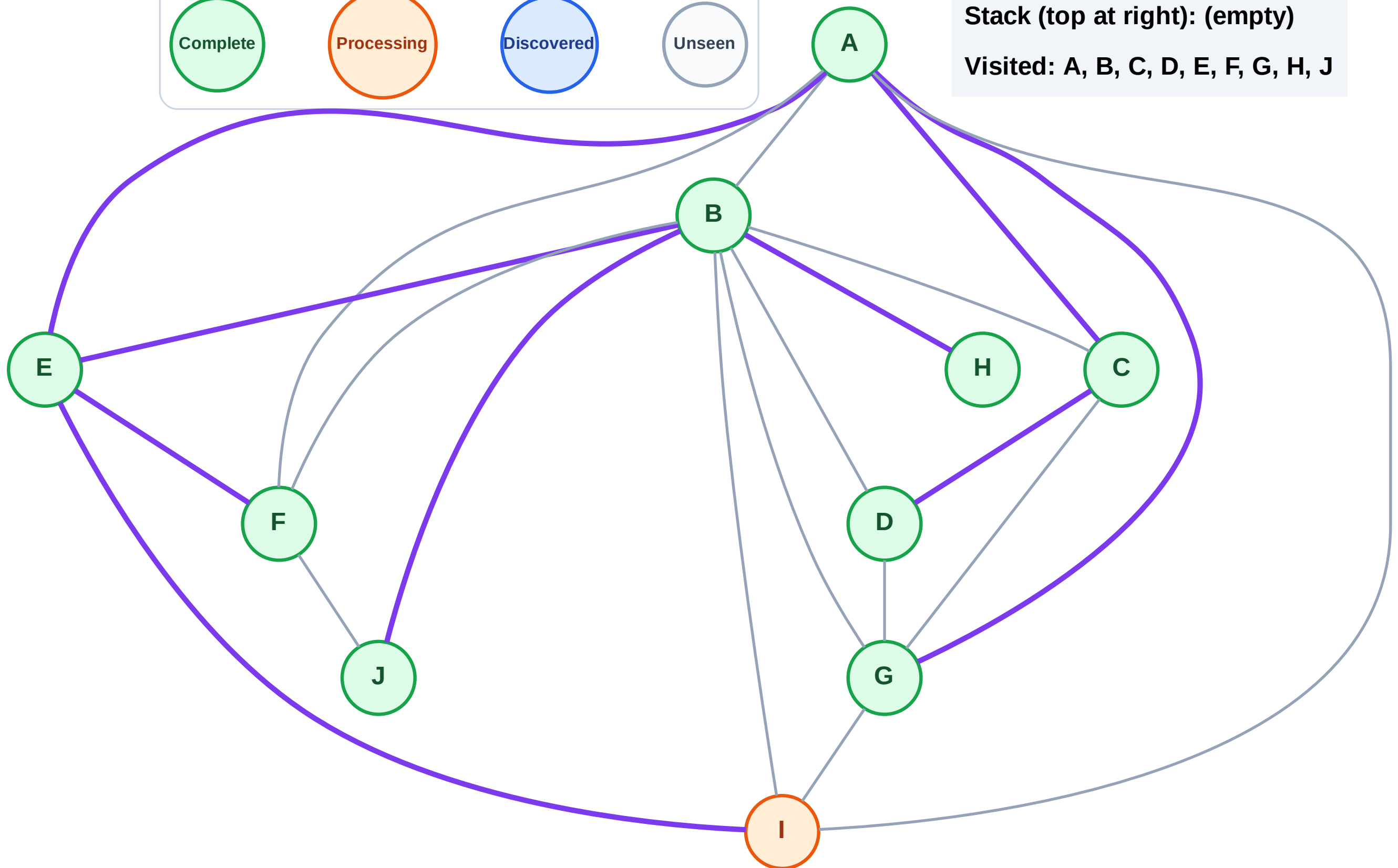
Step 20: Process node I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, J



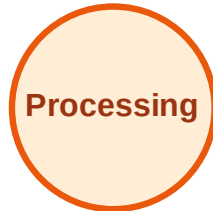
DFS Traversal

Step 21: No new neighbors from I

Legend



Complete



Processing



Discovered



Unseen

Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

